Theoretical Work on Clustering in Leading Statistics and Machine-Learning Journals, 2020–2026

An annotated bibliography, journal by journal and in reverse chronological order, with special attention to results that determine the number of clusters

The Annals of Statistics
Journal of the American Statistical Association
Journal of the Royal Statistical Society, Series B
Biometrika
Journal of Machine Learning Research

Literature checked up to 7 October 2026

Tags. **[#K]** result about the *number* of clusters, components or communities **[net]** network community detection **[group]** grouping / homogeneity pursuit (clustering of parameters). Pre-2020 landmark papers close each journal's section.

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About this bibliography

Scope

The list covers **theoretical (mathematical) articles on clustering** published from 2020 to 2026 in five journals: *The Annals of Statistics* (AoS), the *Journal of the American Statistical Association* (JASA), the *Journal of the Royal Statistical Society, Series B* (JRSS-B), *Biometrika*, and the *Journal of Machine Learning Research* (JMLR). An article is included when its main contribution contains theorems about a clustering problem. Such theorems include consistency, misclustering rates, minimax bounds, recovery thresholds and statistical–computational limits, identifiability, limiting null distributions of tests, posterior concentration, distributional properties of random-partition priors, and approximation guarantees. The clustering problem may concern

- observations: k-means and its relatives, spectral, SDP and convex clustering, density, mode and level-set clustering, hierarchical, subspace and tensor clustering, mixture- and latent-class-based clustering, and Bayesian random partitions;
- variables (variable clustering) or parameters (**[group]**);
- nodes of a network (community detection, flagged **[net]**, included as requested).

Articles whose main subject is the *number* of clusters are flagged **[#K]**. This includes estimating, testing or selecting K; procedures that provably recover the partition without being given K; and results on how a prior or posterior behaves on the number of clusters.

Excluded are articles with no substantive theory, and mixture-model articles that only estimate the mixture density or its parameters, with no clustering or order-selection component. Supervised classification and papers that use clustering only as a tool are also excluded. Some borderline exclusions, after reading the abstract or paper: *Bayesian distance clustering* (Duan & Dunson, JMLR 2021; no formal theory), *Distributed kernel-driven data clustering* (Schizas, JMLR 2024; algorithmic), and *Byzantine-tolerant distributed learning of finite mixture models* (Zhang, Tan & Chen, JRSS-B advance article 2026; parameter estimation).

Organisation

Journals appear in the order AoS, JASA, JRSS-B, Biometrika, JMLR, and each journal is finished before the next begins. Within a journal, years run from 2026 back to 2020. Within a year, articles run from the latest issue (or paper number) to the earliest; advance or online-first articles come first. A short section of *selected landmark papers before 2020*, also in reverse chronological order, ends each journal. Every entry gives the title, tags, authors (year) and volume(issue) with pages, article number or DOI, followed by a short summary of the main theoretical result. Entries are numbered A1, A2, … (AoS), J (JASA), R (JRSS-B), B (Biometrika) and

M (JMLR). Appendix A collects every [#K] entry, and Appendix B closes the document with ten candidate theoretical research projects suggested by the literature, each checked for novelty against existing work.

Sources and caveats

- **How the summaries were written.** Each summary is based on the published abstract or the arXiv version, and on the full text where it was accessible. The summaries are deliberately brief, and the papers themselves should be read for the precise assumptions.
- **How the articles were found.** The tables of contents were collected from publisher pages (Oxford Academic for JRSS-B and Biometrika), RePEc/IDEAS for JASA, jmlr.org and the ML Anthology for JMLR, a bibliographic index of AoS, and targeted searches.
- **AoS.** Enumeration is complete through vol. 52(5) (October 2024). The AoS website could not be read automatically, so items from 52(6) to 54(5) (December 2024 to October 2026) were found only by targeted searches, and that period *may be incomplete*. For those items the DOI is given in place of page numbers.
- **JASA.** Enumeration covers issues through 121(554) (June 2026). Issue 119(545) (March 2024) was missing from the index, so only one article from it is included. One accepted online-first article from 2026 is also included, but later online-first articles were not checked systematically.
- **JRSS-B and Biometrika.** All issues and advance articles available on 6–7 October 2026 were checked.
- **JMLR.** Volumes 21–27 were checked. A few long listing pages were truncated, so isolated omissions in 2022–2024 are possible.

At a glance

Journal	2020–2026	[#K]	[net]	[group]	Landmarks (<2020)
The Annals of Statistics (§1)	39	12	12	1	24
J. Amer. Statist. Assoc. (§2)	45	12	22	1	9
J. R. Stat. Soc. Ser. B (§3)	19	2	8	3	9
Biometrika (§4)	13	5	7	0	4
J. Mach. Learn. Res. (§5)	72	9	25	2	6
Total	**188**	**40**	**74**	**7**	**52**

Tag counts refer to the 2020–2026 entries. Appendix A lists all 71 [#K] entries, including the landmark papers.

Where the theory on the number of clusters is

The [#K] results fall into six families (details in Appendix A):

1. **Tests and goodness of fit, applied sequentially.**
   - Networks: random-matrix and residual statistics (A46, R23, J40, A9, A10, B2); cycle and polygon counts (A21, J32, A20); modularity (B10).
   - Mixtures: EM tests (A54, J50, J49).
   - Classical tests: the dip test (A61), Silverman's test (R28) and SigClust (J51).
2. **Information criteria and penalised likelihood.**
   - SBM criteria: A43, J43.
   - Singular BIC: R22.
   - Group-structure models in time series, panels and point processes: A2, M4, J25, J42.
   - Penalised fusion of mixture atoms: A23.
3. **Spectral and rank methods.**
   - Bethe Hessian and regularised Laplacians: J20, M48.
   - Eigengaps: M72.
   - Ranks of integral operators for nonparametric mixtures and HMMs: A27, M12, R24.
4. **Cross-validation, stability and distortion curves.**

- Network cross-validation: J47, B12.
- Instability and cross-validation: B16.
- Gap and jump statistics: R27, J53.
- The first consistency proof for the elbow method: A4.

5. **Bayesian approaches.**
   - Dirichlet- and Pitman–Yor-process mixtures are inconsistent for K (M77).
   - Consistency is restored by a prior on the number of components (J48, A13), a prior on the concentration (B7), or modified stick-breaking (M28).
   - Overfitted and repulsive or non-local mixtures: R25, J44, R20.
6. **Partition recovered without K as input.**
   - Sparse off-the-grid estimation: A29.
   - Variable clustering: A39, A35, J12.
   - Convex or fusion paths: R21, M15.
   - Density level sets: A47, M26.

1 The Annals of Statistics

Coverage note. Complete through vol. 52(5) (October 2024). Items from 52(6)–54(5) were found by targeted searches only, and this period may be incomplete; for those items the DOI is given in place of pages.

2026

A1 Analysis of singular subspaces under random perturbations

K. Wang (2026). *Ann. Statist. 54(2); doi:10.1214/25-AOS2582*

Extends the Davis–Kahan–Wedin $\sin\Theta$ theorem, for a low-rank signal plus Gaussian noise, to every unitarily invariant norm and adds entrywise bounds. As an application, a spectral algorithm for clustering Gaussian mixtures has a polynomially small expected misclassification rate under separation conditions.

A2 Estimation of grouped time-varying network vector autoregressive models [#K]

D. Li, B. Peng, S. Tang, W. Wu (2026). *Ann. Statist. 54(2); doi:10.1214/25-AOS2580*

Nodes of a network autoregression have smooth time-varying coefficient functions that are shared within latent groups. The groups are recovered by agglomerative clustering on a normalised distance between nonparametric estimates, and their number is chosen by an information criterion. Both the memberships and the number of groups are estimated consistently, and the post-grouping estimators are asymptotically normal.

2025

A3 Clustering by hill-climbing: consistency results

E. Arias-Castro, W. Qiao (2025). *Ann. Statist. 53(6); doi:10.1214/25-AOS2554*

Studies Fukunaga–Hostetler-type hill-climbing (mode-seeking) clustering: mean shift and related continuous-space schemes, and discrete medoid-type variants. Each is shown to be consistent, in the sense that sample points are grouped as the population gradient flow groups them, i.e., by the basins of attraction of the density modes, so the number of clusters is not a tuning input.

A4 Asymptotic theory of geometric and adaptive k-means clustering [#K]

A. Q. Jaffe (2025). *Ann. Statist. 53(4); doi:10.1214/25-AOS2514*

A unified asymptotic theory for k-means-type procedures (including k-medians and k-medoids) in general metric and geometric spaces, in which the domain of the centres and the number of centres may be chosen adaptively. Empirical centre sets are strongly consistent (Kuratowski, and under conditions Hausdorff, convergence) assuming only moments. In particular, choosing k by the elbow method is shown to be consistent, the first such guarantee.

2024

A5 Computational and statistical thresholds in multi-layer stochastic block models [net]

J. Lei, A. R. Zhang, Z. Zhu (2024). *Ann. Statist. 52(5), 2431–2455*

For a two-block multilayer SBM, the density threshold for consistent community recovery improves linearly in the number of layers L when computation is unrestricted, but only like $\sqrt{L}$ for low-degree polynomial algorithms. This reveals a statistical–computational gap that is absent for single-layer networks, and bears on the optimality of bias-adjusted spectral clustering.

A6 Leave-one-out singular subspace perturbation analysis for spectral clustering

A. Y. Zhang, H. H. Zhou (2024). *Ann. Statist. 52(5), 2004–2033*

Leave-one-out perturbation bounds for singular subspaces, sharper than Wedin-type bounds at the level of individual rows. As a consequence, spectral clustering achieves an exponentially small misclustering rate in sub-Gaussian mixture models.

A7 Fundamental limits of low-rank matrix estimation with diverging aspect ratios

A. Montanari, Y. Wu (2024). *Ann. Statist. 52(4), 1460–1484*

Characterises the Bayes-optimal error of estimating the factors of a noisy low-rank $n \times d$ matrix when $d/n \to \infty$ (or 0). Clustering a Gaussian mixture with known equal covariances is a leading example, and there is a regime where the cluster labels can be partially recovered although the centres cannot.

A8 Supervised homogeneity fusion: a combinatorial approach [group]
W. Wang, S. Wu, Z. Zhu, L. Zhou, P. X.-K. Song (2024). *Ann. Statist. 52(1), 285–310*
L_0-Fusion clusters regression coefficients into groups with a common value, using an L_0 (combinatorial) constraint solved by mixed-integer optimisation. The difficulty of the problem is governed by a "grouping sensitivity", and the method is grouping-consistent under the weakest possible condition on it, since below that condition the minimax risk of group misspecification does not vanish. It remains efficient after sure screening in high dimensions.

2023

A9 Adjusted chi-square test for degree-corrected block models [net] [#K]
L. Zhang, A. A. Amini (2023). *Ann. Statist. 51(6), 2366–2385*
Goodness-of-fit test for DCSBMs, built on an adjusted chi-square statistic applied to a degree-conditional compression of the adjacency matrix. The statistic has a valid null limit whenever the harmonic mean of the degrees diverges. Applied sequentially, with rows compressed by a $(K+1)$-community fit, it recovers the number of communities consistently, and it is consistent against general latent-variable alternatives.

A10 Universal rank inference via residual subsampling with application to large networks [net] [#K]
X. Han, Q. Yang, Y. Fan (2023). *Ann. Statist. 51(3), 1109–1133*
RIRS tests the rank of the mean matrix of a large network, e.g. the number of communities in SBM, DCSBM or mixed-membership models. The statistic is built from subsampled residual entries and is asymptotically standard normal under the null for a broad family of models. Sequential testing then estimates K consistently.

A11 Optimal estimation and computational limit of low-rank Gaussian mixtures
Z. Lyu, D. Xia (2023). *Ann. Statist. 51(2), 646–667*
Low-rank Gaussian mixture model for matrix-valued observations (e.g. clustering multi-layer networks or images). Establishes minimax rates, attained by the MLE at the statistical limit and by a fast spectral-aggregation estimator above the computational limit. Low-degree evidence indicates that no polynomial-time algorithm succeeds in between, so the problem shows multiple phase transitions and a statistical–computational gap.

2022

A12 Global and individualized community detection in inhomogeneous multilayer networks [net]
S. Chen, S. Liu, Z. Ma (2022). *Ann. Statist. 50(5), 2664–2693*
Each layer follows its own SBM, and its communities are random perturbations of a common global partition. Sharp-exponent minimax rates are derived both for the global partition and for layer-wise (individualised) memberships, and one efficient algorithm attains both. The rates depend on the parity of the number of most informative layers.

A13 Is infinity that far? A Bayesian nonparametric perspective of finite mixture models [#K]
R. Argiento, M. De Iorio (2022). *Ann. Statist. 50(5), 2641–2663*
Mixtures of finite mixtures with normalised independent finite point processes as mixing priors. The exchangeable partition probability function, the prior and posterior of the number of components and of the number of clusters, and predictive (urn) schemes are all derived in closed form. This places finite mixtures within the analytic toolbox of Bayesian nonparametrics.

A14 An ℓ_p theory of PCA and spectral clustering
E. Abbe, J. Fan, K. Wang (2022). *Ann. Statist. 50(4), 2359–2385*
ℓ_p perturbation theory for eigenvectors of hollowed Gram matrices, proved by leave-one-out arguments. Vanilla spectral clustering, with no refinement step, achieves optimal exact-recovery thresholds and misclassification rates in sub-Gaussian mixtures and in contextual community detection.

A15 Sharp optimal recovery in the two component Gaussian mixture model
M. Ndaoud (2022). *Ann. Statist. 50(4), 2096–2126*
Gives the sharp separation threshold, which depends on dimension, noise and sample size, for exact label recovery in the symmetric two-component Gaussian mixture with unknown centres, including

$p \gg n$. A polynomial-time iterative algorithm attains it, along with optimal Hamming-loss rates.

A16 Iterative algorithm for discrete structure recovery
C. Gao, A. Y. Zhang (2022). *Ann. Statist. 50(2), 1066–1094*
A general framework for Lloyd- and EM-type iterative algorithms that recover discrete structures such as labels. Under a one-step contraction condition and a reasonable initialisation, they attain minimax-optimal misclassification. Applications include Gaussian mixture clustering, community detection and approximate ranking.

A17 Tensor clustering with planted structures: statistical optimality and computational limits
Y. Luo, A. R. Zhang (2022). *Ann. Statist. 50(1), 584–613*
High-order clustering, i.e., detection and recovery of planted sub-tensor (tensor biclustering) structures. Information-theoretic thresholds are established, together with computational thresholds based on hypergraphic planted clique hardness. Together they give a complete phase diagram with statistical–computational gaps.

A18 Semiparametric latent-class models for multivariate longitudinal and survival data
K. Y. Wong, D. Zeng, D. Y. Lin (2022). *Ann. Statist. 50(1), 487–510*
Latent classes jointly drive multivariate longitudinal outcomes and event times, with nonparametric baseline components. The paper establishes identifiability, consistency, asymptotic normality and semiparametric efficiency of the nonparametric MLE.

A19 Testing equivalence of clustering
C. Gao, Z. Ma (2022). *Ann. Statist. 50(1), 407–429*
Tests whether two data sets (e.g. two Gaussian mixtures, or two networks on the same subjects) share the same clustering structure. Minimax separation rates, measured in a Hamming-type distance between the label vectors, are derived, and optimal tests are constructed in different regimes.

A20 Testing community structure for hypergraphs [net] [#K]
M. Yuan, R. Liu, Y. Feng, Z. Shang (2022). *Ann. Statist. 50(1), 147–169*
When can a uniform hypergraph with communities be told apart from its Erdős–Rényi counterpart, i.e., $K > 1$ from $K = 1$? The paper finds a phase transition in the hyperedge probability and sharp SNR thresholds in the bounded-degree regime. In the dense regime it gives sub-hypergraph-count tests with explicit null distributions.

2021

A21 Optimal adaptivity of signed-polygon statistics for network testing [net] [#K]
J. Jin, Z. T. Ke, S. Luo (2021). *Ann. Statist. 49(6), 3408–3433*
Global test of $K = 1$ against $K > 1$ communities under a degree-corrected mixed-membership model with severe degree heterogeneity. Signed-polygon (signed triangle and quadrilateral) statistics have parameter-free $N(0, 1)$ nulls and are optimally adaptive, matching the lower bounds across sparsity and heterogeneity regimes.

A22 Community detection on mixture multilayer networks via regularized tensor decomposition [net]
B.-Y. Jing, T. Li, Z. Lyu, D. Xia (2021). *Ann. Statist. 49(6), 3181–3205*
Layers come from a few different SBMs. TWIST, a regularised tensor (Tucker) decomposition, clusters both the layers and the nodes, with error bounds for global and local memberships that hold in sparse regimes.

A23 Estimating the number of components in finite mixture models via the Group-Sort-Fuse procedure [#K]
T. Manole, A. Khalili (2021). *Ann. Statist. 49(6), 3043–3069*
Fits an overfitted mixture with a penalty that fuses sorted, grouped atoms. The procedure estimates the number of components consistently and the mixing measure at optimal rates for strongly identifiable families, including multivariate ones.

A24 Optimality of spectral clustering in the Gaussian mixture model
M. Löffler, A. Y. Zhang, H. H. Zhou (2021). *Ann. Statist. 49(5), 2506–2530*

Vanilla spectral clustering (SVD followed by k-means) attains the minimax misclustering rate $\exp\{-(1+o(1))\Delta^2/8\}$ for isotropic Gaussian mixtures, with no refinement step. The proof uses an entrywise (leave-one-out) analysis of the singular vectors.

A25 Community detection with dependent connectivity [net]

Y. Yuan, A. Qu (2021). *Ann. Statist. 49(4), 2378–2428*

Lets edges be dependent and models the dependence through a Bahadur representation in a likelihood-based approach. Community estimates are consistent, and exploiting the dependence gives faster convergence than methods that assume independent edges.

A26 Robust k-means clustering for distributions with two moments

Y. Klochkov, A. Kroshnin, N. Zhivotovskiy (2021). *Ann. Statist. 49(4), 2206–2230*

A median-of-means-type k-means procedure with non-asymptotic excess-distortion bounds, and fast rates under a margin condition, assuming only two finite moments. The bounds also hold under adversarial contamination.

A27 Estimation of the number of components of nonparametric multivariate finite mixture models [#K]

C. Kwon, E. Mbakop (2021). *Ann. Statist. 49(4), 2178–2205*

In nonparametric mixtures with conditionally independent coordinates, the number of components equals the rank of an integral operator. Thresholding the singular values of its estimate gives consistent estimators of K and of lower bounds on it.

A28 Nonexchangeable random partition models for microclustering

G. Di Benedetto, F. Caron, Y. W. Teh (2021). *Ann. Statist. 49(4), 1931–1957*

Random partition models driven by the arrival order of data, so that cluster sizes grow sublinearly (the microclustering property needed for entity resolution). Asymptotic properties of the cluster sizes and of the number of clusters are derived.

A29 SuperMix: sparse regularization for mixtures [#K]

Y. De Castro, S. Gadat, C. Marteau, C. Maugis-Rabusseau (2021). *Ann. Statist. 49(3), 1779–1809*

Off-the-grid Beurling-LASSO estimation of a mixing measure from kernel or characteristic-function moments. With high probability it recovers exactly the number of mixture components, and it estimates locations and weights at near-parametric rates under separation conditions.

A30 Robust Bregman clustering

C. Brécheteau, A. Fischer, C. Levrard (2021). *Ann. Statist. 49(3), 1679–1701*

Trimmed k-means generalised to Bregman divergences. The paper proves existence and consistency of trimmed optimal codebooks, non-asymptotic convergence rates for the trimmed distortion, and robustness (breakdown) guarantees against outliers.

A31 Concentration of kernel matrices with application to kernel spectral clustering

A. A. Amini, Z. S. Razaee (2021). *Ann. Statist. 49(1), 531–556*

Proves operator-norm concentration of random kernel matrices around their population counterparts. As a consequence, kernel spectral clustering under nonparametric mixture models has explicit misclassification bounds and is consistent.

2020

A32 Clustering in block Markov chains

J. Sanders, A. Proutière, S.-Y. Yun (2020). *Ann. Statist. 48(6), 3488–3512*

A single trajectory of a Markov chain is observed, and its transition kernel depends only on hidden clusters of states. The paper gives an information-theoretic lower bound on the number of misclassified states, and shows that spectral clustering followed by likelihood-based improvement attains it asymptotically.

A33 Theoretical and computational guarantees of mean field variational inference for community detection [net]

A. Y. Zhang, H. H. Zhou (2020). *Ann. Statist. 48(5), 2575–2598*

From a suitable initialisation, batch coordinate-ascent mean-field variational inference for the SBM converges linearly and attains the minimax-optimal misclassification rate.

A34 Identifiability of nonparametric mixture models and Bayes optimal clustering
B. Aragam, C. Dan, E. P. Xing, P. Ravikumar (2020). *Ann. Statist. 48(4), 2277–2302*
Nonparametric mixtures are identifiable when the components are suitably separated, and this is shown through a clustering of the mixture's density. The paper defines the Bayes-optimal partition for nonparametric mixtures and gives consistent estimators of it.

A35 Adaptive estimation in structured factor models with applications to overlapping clustering [#K]
X. Bing, F. Bunea, Y. Ning, M. Wegkamp (2020). *Ann. Statist. 48(4), 2055–2081*
LOVE estimates a sparse latent factor model with pure variables. It is proved identifiable, it estimates the number of latent clusters K consistently, and it recovers the pure variables and the overlapping cluster memberships with minimax-optimal rates.

A36 Entrywise eigenvector analysis of random matrices with low expected rank [net]
E. Abbe, J. Fan, K. Wang, Y. Zhong (2020). *Ann. Statist. 48(3), 1452–1474*
First-order ℓ_∞ approximation $u \approx Au^*/\lambda^*$ for eigenvectors of random matrices with low expected rank. Consequences include that vanilla spectral algorithms achieve the exact-recovery threshold in the SBM, as well as results for $\mathbb{Z}_2$-synchronisation and matrix completion.

A37 Spectral and matrix factorization methods for consistent community detection in multi-layer networks [net]
S. Paul, Y. Chen (2020). *Ann. Statist. 48(1), 230–250*
Multilayer SBMs with common communities. Gives misclustering bounds for spectral clustering on aggregate matrices and for orthogonal linked matrix factorisation, and these bounds improve as the number of layers grows.

A38 Optimal rates for community estimation in the weighted stochastic block model [net]
M. Xu, V. Jog, P.-L. Loh (2020). *Ann. Statist. 48(1), 183–204*
SBMs with general edge-weight distributions. The minimax misclassification rate is governed by a Rényi divergence of order $1/2$ between the within- and between-community weight distributions, and an adaptive algorithm attains it without knowing those distributions.

A39 Model assisted variable clustering: minimax-optimal recovery and algorithms [#K]
F. Bunea, C. Giraud, X. Luo, M. Royer, N. Verzelen (2020). *Ann. Statist. 48(1), 111–137*
G-block covariance models for clustering variables. The paper derives the minimax separation for exact recovery, attains it with PECOK (a corrected SDP relaxation of k-means), and gives COD, an algorithm that recovers the partition without being given the number of groups.

Selected landmark papers before 2020

A40 CHIME: clustering of high-dimensional Gaussian mixtures with EM algorithm and its optimality
T. T. Cai, J. Ma, L. Zhang (2019). *Ann. Statist. 47(3), 1234–1267*
A sparse high-dimensional EM algorithm (estimating the discriminant direction) that achieves optimal rates both for the parameters and for the misclustering error.

A41 Community detection in degree-corrected block models [net]
C. Gao, Z. Ma, A. Y. Zhang, H. H. Zhou (2018). *Ann. Statist. 46(5), 2153–2185*
Minimax misclassification rates for DCSBMs, which reflect degree heterogeneity, and a polynomial-time algorithm that attains them.

A42 Phase transitions for high dimensional clustering and related problems
J. Jin, Z. T. Ke, W. Wang (2017). *Ann. Statist. 45(5), 2151–2189*
Sharp phase diagrams for detecting and recovering two clusters in sparse high-dimensional mixtures, distinguishing statistically possible regions from computationally tractable ones.

A43 Likelihood-based model selection for stochastic block models [net] [#K]
Y. X. R. Wang, P. J. Bickel (2017). *Ann. Statist. 45(2), 500–528*

Asymptotics of the log-likelihood ratio between SBMs with different numbers of blocks, and a penalised-likelihood criterion that estimates K consistently.

A44 Influential features PCA for high dimensional clustering

J. Jin, W. Wang (2016). *Ann. Statist. 44(6), 2323–2359*

IF-PCA (discussion paper) selects features by Kolmogorov–Smirnov screening and then applies PCA. It is proved consistent, and phase transitions for the clustering error are characterised.

A45 Minimax rates of community detection in stochastic block models [net]

A. Y. Zhang, H. H. Zhou (2016). *Ann. Statist. 44(5), 2252–2280*

The minimax misclassification proportion in the SBM decays exponentially in nI/K, where I is the Rényi divergence of order $1/2$ between the edge distributions. This benchmark is used throughout the later literature.

A46 A goodness-of-fit test for stochastic block models [net] [#K]

J. Lei (2016). *Ann. Statist. 44(1), 401–424*

The largest singular value of the standardised residual matrix has a Tracy–Widom limit under the null. Testing sequentially gives a consistent estimate of K.

A47 Fully adaptive density-based clustering [#K]

I. Steinwart (2015). *Ann. Statist. 43(5), 2132–2167*

Estimates the first level at which the density's level sets split, together with the resulting clusters (connected components), adaptively and at optimal rates. The number of clusters is therefore determined, not prescribed.

A48 Consistency of spectral clustering in stochastic block models [net]

J. Lei, A. Rinaldo (2015). *Ann. Statist. 43(1), 215–237*

Misclustering bounds for adjacency spectral clustering in sparse SBMs and DCSBMs, obtained from sharp spectral-norm concentration.

A49 Fast community detection by SCORE [net]

J. Jin (2015). *Ann. Statist. 43(1), 57–89*

SCORE takes entrywise ratios of the leading eigenvectors to remove degree heterogeneity in DCSBMs, and comes with consistency guarantees.

A50 Consistency of community detection in networks under degree-corrected stochastic block models [net]

Y. Zhao, E. Levina, J. Zhu (2012). *Ann. Statist. 40(4), 2266–2292*

A general theory of strong and weak consistency of criterion-based community detection (likelihood and modularity) under DCSBMs.

A51 A geometric analysis of subspace clustering with outliers

M. Soltanolkotabi, E. J. Candès (2012). *Ann. Statist. 40(4), 2195–2238*

Geometric conditions under which sparse subspace clustering correctly recovers subspaces that may intersect, and detects outliers.

A52 Spectral clustering and the high-dimensional stochastic blockmodel [net]

K. Rohe, S. Chatterjee, B. Yu (2011). *Ann. Statist. 39(4), 1878–1915*

The first high-dimensional consistency analysis of normalised-Laplacian spectral clustering in SBMs whose number of blocks grows.

A53 Generalized density clustering

A. Rinaldo, L. Wasserman (2010). *Ann. Statist. 38(5), 2678–2722*

Consistency and rates for estimating density level-set clusters and the cluster tree from kernel density estimates.

A54 Hypothesis test for normal mixture models: the EM approach [#K]

J. Chen, P. Li (2009). *Ann. Statist. 37(5A), 2523–2542*

The EM test for the order of normal mixtures has a simple, tractable limiting distribution, which avoids the irregularity of the likelihood-ratio test.

A55 Consistency of spectral clustering

U. von Luxburg, M. Belkin, O. Bousquet (2008). *Ann. Statist. 36(2), 555–586*

Normalised spectral clustering is consistent as the sample size grows, whereas unnormalised spectral clustering can fail.

A56 On the posterior distribution of the number of components in a finite mixture [#K]

A. Nobile (2004). *Ann. Statist. 32(5), 2044–2073*

Derives the posterior distribution of the number of components and studies its asymptotic behaviour.

A57 Consistent estimation of mixture complexity [#K]

L. F. James, C. E. Priebe, D. J. Marchette (2001). *Ann. Statist. 29(5), 1281–1296*

A semiparametric estimator of the number of components, based on comparing a kernel density estimate with a fitted mixture, is shown to be consistent.

A58 Testing the order of a model using locally conic parametrization: population mixtures and stationary ARMA processes [#K]

D. Dacunha-Castelle, É. Gassiat (1999). *Ann. Statist. 27(4), 1178–1209*

Locally conic parametrisation yields the limiting distribution of likelihood-ratio tests for the order of a mixture, and consistent estimators of that order.

A59 Measuring mass concentrations and estimating density contour clusters—an excess mass approach [#K]

W. Polonik (1995). *Ann. Statist. 23(3), 855–881*

The excess-mass functional is used to estimate density contour clusters and to test for multimodality.

A60 Consistent estimation of a mixing distribution [#K]

B. G. Leroux (1992). *Ann. Statist. 20(3), 1350–1360*

Maximum-penalised-likelihood (AIC/BIC-type) estimators of a finite mixing distribution are consistent, and asymptotically they do not underestimate the number of components.

A61 The dip test of unimodality [#K]

J. A. Hartigan, P. M. Hartigan (1985). *Ann. Statist. 13(1), 70–84*

The dip statistic (distance to the closest unimodal distribution) and its null theory, which give a test of one cluster against several.

A62 Strong consistency of k-means clustering

D. Pollard (1981). *Ann. Statist. 9(1), 135–140*

The optimal empirical k-means centres converge almost surely to the population optimum.

A63 Asymptotic distributions for clustering criteria [#K]

J. A. Hartigan (1978). *Ann. Statist. 6(1), 117–131*

Asymptotic distribution of the optimal split (two-means) criterion, the basis for testing whether a population divides into two clusters.

2 Journal of the American Statistical Association

Coverage note. Theory and Methods articles through vol. 121(554) (June 2026), plus one accepted online-first article. Only one article could be identified from issue 119(545).

2026

J1 Scalable community detection in massive networks via predictive assignment [net]
S. Bhadra, M. Pensky, S. Sengupta (2026). *J. Amer. Statist. Assoc., accepted (online 2026); doi:10.1080/01621459.2026.2628341*
Community detection is run on a small random subgraph, and the remaining nodes are then assigned in parallel by a predictive rule. Strong consistency holds under SBMs and DCSBMs even when the clustering of the subgraph is only weakly consistent.

J2 When does bottom-up beat top-down in hierarchical community detection? [net]
M. Dreveton, D. Kuroda, M. Grossglauser, P. Thiran (2026). *J. Amer. Statist. Assoc. 121(554), 1284–1295*
Guarantees for recovering the tree and the communities of a hierarchical SBM with a bottom-up (agglomerative) algorithm, which attains the information-theoretic threshold for exact recovery at intermediate levels of the hierarchy. These conditions are weaker than those known for top-down (divisive) algorithms.

J3 A new approach for homogeneity pursuit in short panel data analysis [group]
Y. Han, W. Wu, W. Zhang (2026). *J. Amer. Statist. Assoc. 121(554), 1511–1521*
Recovers latent groups of individuals that share covariate effects in panel data. Unlike earlier methods, which need large clusters or long panels, the approach also works for short panels, and asymptotic properties of the estimated grouping and coefficients are established.

J4 Deep clustering evaluation: how to validate internal clustering validation measures
Z. Wang, C. Ye (2026). *J. Amer. Statist. Assoc. 121(553), 223–236*
(Partly theoretical.) Shows theoretically why traditional internal validation indices mislead in deep clustering: they fail on high-dimensional inputs, and they are not comparable across different learned embedding spaces. The paper then aggregates the indices over embedding spaces whose rankings of candidate partitions agree.

2025

J5 Degree-heterogeneous latent class analysis for high-dimensional discrete data
Z. Lyu, L. Chen, Y. Gu (2025). *J. Amer. Statist. Assoc. 120(552), 2435–2448*
Latent class models with subject-specific degree heterogeneity, for high-dimensional binary or categorical responses. A spectral clustering method (HeteroClustering) recovers the latent classes exactly with high probability as the dimension grows, and the item parameters come with estimation guarantees.

J6 Large precision matrix estimation with unknown group structure
C. Cheng, Y. Ke, W. Zhang (2025). *J. Amer. Statist. Assoc. 120(552), 2326–2337*
Features are clustered in the space of the leading sample eigenvectors to detect an unknown group structure, and group-wise multivariate regressions then estimate the precision matrix. Theory covers both the group detection and the precision-matrix estimation.

J7 Dependent random partitions by shrinking toward an anchor
D. B. Dahl, R. L. Warr, T. P. Jensen (2025). *J. Amer. Statist. Assoc. 120(552), 2684–2696*
The shrinkage partition distribution shrinks any partition distribution toward an anchor partition, with item-specific shrinkage parameters ranging from independence to equality. It has a tractable normalising constant, it is used for hierarchically and temporally dependent partitions, and its formal properties are proved.

J8 Joint spectral clustering in multilayer degree-corrected stochastic blockmodels [net]
J. Agterberg, Z. Lubberts, J. Arroyo (2025). *J. Amer. Statist. Assoc. 120(551), 1607–1620*
Multilayer DCSBMs with common communities but layer-specific connectivity and degrees. The

algorithm aggregates normalised spectral embeddings across layers and provably achieves perfect clustering, via entrywise eigenvector perturbation bounds.

J9 Bayesian clustering via fusing of localized densities

A. Dombowsky, D. B. Dunson (2025). *J. Amer. Statist. Assoc. 120(551), 1775–1786*

FOLD fits an overfitted Bayesian mixture and merges components according to the posterior of their localised kernels, which makes the clusters robust to kernel misspecification and favours a small number of clusters. The method has a decision-theoretic justification and optimal clustering under kernel misspecification.

J10 Partially exchangeable stochastic block models for (node-colored) multilayer networks [net]

D. Durante, F. Gaffi, A. Lijoi, I. Prünster (2025). *J. Amer. Statist. Assoc. 120(551), 1811–1827*

SBMs for node-coloured multilayer networks with a hierarchical random-partition prior, so that the number of groups is learned. The paper derives closed-form predictive co-clustering probabilities within and across layers, and notions of node exchangeability suited to multilayer networks.

J11 Network goodness-of-fit for the block-model family [net]

J. Jin, Z. T. Ke, J. Tang, J. Wang (2025). *J. Amer. Statist. Assoc. 120(551), 1919–1932*

GoF-MSCORE builds goodness-of-fit metrics for DCMM, and by extension for SBM, DCBM and MMSBM, from cycle counts after a network-fitting step. Each metric has a parameter-free $N(0,1)$ limit when the assumed model is correct and is power-optimal in many settings. DCMM is shown to be nearly as broad as the rank-K network model.

J12 High-dimensional variable clustering based on maxima of a weakly dependent random process [#K]

A. Boulin, E. Di Bernardino, T. Laloë, G. Toulemonde (2025). *J. Amer. Statist. Assoc. 120(551), 1933–1944*

Variable clustering for extremes under an asymptotically-independent-block model, in which clusters of coordinates are independent in their extremes. The algorithm, based on extremal dependence, needs no prespecified number of clusters, and it recovers the partition exactly with high probability in high dimensions for weakly dependent processes.

J13 Node-level community detection within edge exchangeable models for interaction processes [net]

Y. Zhang, W. Dempsey (2025). *J. Amer. Statist. Assoc. 120(550), 764–778*

Block edge-exchangeable models, such as the block vertex components model, for interaction processes with latent node communities. They allow sparse, power-law degrees within communities, and theory bounds the misspecification rate of the block assignments.

J14 Optimal network membership estimation under severe degree heterogeneity [net]

Z. T. Ke, J. Wang (2025). *J. Amer. Statist. Assoc. 120(550), 948–962*

Under DCMM, the optimal rate of mixed-membership estimation is an explicit functional of the degree-heterogeneity distribution. Mixed-SCORE with a pre-PCA normalisation by $D^{-1/2}$ is rate-optimal for arbitrary heterogeneity, as shown by entrywise eigenvector analysis.

J15 Estimating higher-order mixed memberships via the $\ell_{2,\infty}$ tensor perturbation bound [net]

J. Agterberg, A. R. Zhang (2025). *J. Amer. Statist. Assoc. 120(550), 1214–1224*

Tensor mixed-membership blockmodel. The paper proves identifiability and per-node error bounds for HOOI combined with simplex corner-finding, using a new leave-one-out $\ell_{2,\infty}$ perturbation bound for tensor SVD under heteroskedastic noise.

J16 Graph-aligned random partition model (GARP)

G. Rebaudo, P. Müller (2025). *J. Amer. Statist. Assoc. 120(549), 486–497*

GARP is a Gibbs-type random partition prior whose clusters are aligned on a graph (e.g. cell lineages). It has an analytic probability mass function, a generalised Chinese-restaurant representation, and an efficient MCMC sampler.

2024

J17 Clustering high-dimensional noisy categorical data

Z. Tian, J. Xu, J. Tang (2024). *J. Amer. Statist. Assoc. 119(548), 3008–3019*

With m attributes, n subjects, r clusters and missingness ϵ, a spectral algorithm recovers the clusters exactly when $mn(1-\epsilon) \gtrsim r^2 \log^3 M$, where $M = \max(m, n)$. A nearly matching necessary condition holds for any algorithm.

J18 Consistent community detection in inter-layer dependent multi-layer networks [net]

J. Zhang, J. Wang, X. Wang (2024). *J. Amer. Statist. Assoc. 119(548), 3141–3151*

Multilayer networks whose layers are dependent rather than independent. The dependence is modelled explicitly, and the estimated community memberships are proved consistent.

J19 Dissecting gene expression heterogeneity: generalized Pearson correlation squares and the K-lines clustering algorithm

J. J. Li, H. J. Zhou, P. J. Bickel, X. Tong (2024). *J. Amer. Statist. Assoc. 119(548), 2450–2463*

Generalised Pearson correlation squares capture mixtures of linear relationships between variables, and the paper derives their asymptotic distributions for population-level inference. K-lines clustering detects the linear components, with K chosen data-adaptively.

J20 On the estimation of the number of communities for sparse networks [net] [#K]

N. Hwang, J. Xu, S. Chatterjee, S. Bhattacharyya (2024). *J. Amer. Statist. Assoc. 119(547), 1895–1910*

Proves that counting the negative eigenvalues of a suitably tuned Bethe Hessian matrix estimates the number of communities consistently in sparse SBMs and DCSBMs.

J21 Multivariate sparse clustering for extremes [#K]

N. Meyer, O. Wintenberger (2024). *J. Amer. Statist. Assoc. 119(547), 1911–1922*

MUSCLE identifies groups of variables that are jointly extreme, using sparse regular variation (Euclidean projection onto the simplex). It is consistent and asymptotically normal, and a score-type criterion selects the number of extremal clusters.

J22 Spectral embedding of weighted graphs [net]

I. Gallagher, A. Jones, A. Bertiger, C. E. Priebe, P. Rubin-Delanchy (2024). *J. Amer. Statist. Assoc. 119(547), 1923–1932*

Asymptotics of spectral embeddings under different edge-weight representations in a generic low-rank model, compared by an information-theoretic measure of how well communities separate. Tempering or thresholding the weights can provably help.

J23 PCABM: pairwise covariates-adjusted block model for community detection [net]

S. Huang, J. Sun, Y. Feng (2024). *J. Amer. Statist. Assoc. 119(547), 2092–2104*

SBMs with pairwise covariates. The MLE of the covariate effects is consistent and asymptotically normal, and spectral clustering with adjustment (SCWA) is consistent for community detection.

J24 Spectral clustering, Bayesian spanning forest, and forest process

L. L. Duan, A. Roy (2024). *J. Amer. Statist. Assoc. 119(547), 2140–2153*

A Bayesian spanning-forest model gives spectral clustering a generative basis. Its posterior connecting matrix has almost the same leading eigenvectors as normalised spectral clustering, and a "forest process" (a graph extension of urn processes) characterises the partition probabilities.

J25 Group network Hawkes process [#K]

G. Fang, G. Xu, H. Xu, X. Zhu, Y. Guan (2024). *J. Amer. Statist. Assoc. 119(547), 2328–2344*

Network Hawkes processes for individuals in latent groups. The paper studies the estimators of group memberships and parameters when the number of groups is over-specified and when it is correct, and gives a data-driven criterion that identifies the true number of groups consistently.

J26 Finite-dimensional discrete random structures and Bayesian clustering

A. Lijoi, I. Prünster, T. Rigon (2024). *J. Amer. Statist. Assoc. 119(546), 929–941*

A general class of finite-dimensional discrete random probability measures, which extends Dirichlet- and Pitman–Yor-multinomial processes, is proposed as a mixing prior. The paper derives partition distributions, the distribution of the number of clusters and posterior characterisations, and links the class to mixtures of finite mixtures.

J27 Skeleton clustering: dimension-free density-aided clustering

Z. Wei, Y.-C. Chen (2024). *J. Amer. Statist. Assoc. 119(546), 1124–1135*

Knots from an overfitted k-means are linked by density-aided similarities and then clustered hierarchically. The similarities are estimated at rates that do not depend on the ambient dimension.

J28 Factor modeling for clustering high-dimensional time series

B. Zhang, G. Pan, Q. Yao, W. Zhou (2024). *J. Amer. Statist. Assoc. 119(546), 1252–1263*

Clusters of time series carry cluster-specific factors in addition to common factors, and some series may belong to no cluster. Consistency with explicit rates is shown for the common factors, the cluster-specific factors and the latent clusters.

J29 Selective inference for hierarchical clustering

L. L. Gao, J. Bien, D. Witten (2024). *J. Amer. Statist. Assoc. 119(545), 332–342*

Tests the difference in means between two clusters estimated by hierarchical clustering, conditioning on the clustering event. The resulting selective p-value controls the selective type I error exactly and is computed efficiently for common linkages.

2023

J30 Bias-adjusted spectral clustering in multi-layer stochastic block models [net]

J. Lei, K. Z. Lin (2023). *J. Amer. Statist. Assoc. 118(544), 2433–2445*

Spectral clustering of the sum of squared adjacency matrices with its diagonal bias removed. As the number of layers grows, it is consistent under much weaker per-layer signal than aggregation-based methods.

J31 Community detection in general hypergraph via graph embedding [net]

Y. Zhen, J. Wang (2023). *J. Amer. Statist. Assoc. 118(543), 1620–1629*

Community detection in non-uniform hypergraphs through a low-dimensional embedding, with asymptotic consistency of the estimated communities.

J32 Optimal estimation of the number of network communities [net] [#K]

J. Jin, Z. T. Ke, S. Luo, M. Wang (2023). *J. Amer. Statist. Assoc. 118(543), 2101–2116*

StGoF fits each candidate K with MSCORE and applies a goodness-of-fit statistic with an $N(0,1)$ limit, stepping up in K. It estimates the number of communities consistently under DCMM with severe degree heterogeneity, and is optimal in a phase-transition sense.

J33 Fast network community detection with profile-pseudo likelihood methods [net]

J. Wang, J. Zhang, B. Liu, J. Zhu, J. Guo (2023). *J. Amer. Statist. Assoc. 118(542), 1359–1372*

A profile-pseudo-likelihood that decouples row and column labels, for SBMs and DCSBMs. The fast alternating algorithm has a convergence guarantee, and the estimated community labels are consistent.

J34 Eigen selection in spectral clustering: a theory-guided practice

X. Han, X. Tong, Y. Fan (2023). *J. Amer. Statist. Assoc. 118(541), 109–121*

The informative eigenvectors for spectral clustering need not be the leading ones. Eigen selection picks them with tests derived from asymptotic theory, and the selection and the resulting clustering are proved consistent.

2022

J35 A doubly enhanced EM algorithm for model-based tensor clustering

Q. Mai, X. Zhang, Y. Pan, K. Deng (2022). *J. Amer. Statist. Assoc. 117(540), 2120–2134*

DEEM is a doubly enhanced EM algorithm for tensor-normal mixtures. It uses sparse discriminant directions in the E-step and structured covariance estimates in the M-step, and attains an optimal clustering-error rate in high dimensions.

J36 Directed community detection with network embedding [net]

J. Zhang, X. He, J. Wang (2022). *J. Amer. Statist. Assoc. 117(540), 1809–1819*

Sender and receiver embeddings for directed networks, which are then clustered. The estimated communities are asymptotically consistent.

J37 Random partition models for microclustering tasks [#K]

B. Betancourt, G. Zanella, R. C. Steorts (2022). *J. Amer. Statist. Assoc. 117(539), 1215–1227*

Exchangeable-sequences-of-clusters (ESC) priors, in which cluster sizes grow sublinearly (microclustering), for entity resolution. The paper characterises the induced partitions, including the asymptotic behaviour of the number of clusters.

J38 Hierarchical community detection by recursive partitioning [net] [#K]

T. Li, L. Lei, S. Bhattacharyya, K. Van den Berge, P. Sarkar, P. J. Bickel, E. Levina (2022). *J. Amer. Statist. Assoc. 117(538), 951–968*

Recursive bipartitioning (e.g. by the sign of the second eigenvector) with a stopping rule that determines the number of communities. Under a binary-tree SBM it recovers the hierarchy and the communities consistently, and it needs weaker conditions than K-way methods at the top levels.

2021

J39 Estimating mixed memberships with sharp eigenvector deviations [net]

X. Mao, P. Sarkar, D. Chakrabarti (2021). *J. Amer. Statist. Assoc. 116(536), 1928–1940*

Mixed-membership SBMs. Sharp row-wise eigenvector deviation bounds give consistent estimation of the membership vectors, with rates, by a simplex corner-finding algorithm.

J40 Using maximum entry-wise deviation to test the goodness of fit for stochastic block models [net] [#K]

J. Hu, J. Zhang, H. Qin, T. Yan, J. Zhu (2021). *J. Amer. Statist. Assoc. 116(535), 1373–1382*

A goodness-of-fit test for SBMs based on the maximum entry-wise deviation of a standardised residual matrix, with an extreme-value null limit. Applying the test sequentially estimates the number of communities.

J41 Covariate regularized community detection in sparse graphs [net]

B. Yan, P. Sarkar (2021). *J. Amer. Statist. Assoc. 116(534), 734–745*

An SDP for community detection that adds a regularisation term built from node covariates. It is consistent even when the graph alone is too sparse, or the covariates alone too weak, for consistent recovery.

J42 Grouped heterogeneous mixture modeling for clustered data [#K]

S. Sugasawa (2021). *J. Amer. Statist. Assoc. 116(534), 999–1010*

The conditional distribution in each cluster is a mixture of latent distributions common to all clusters, and the clusters are grouped by their mixing proportions. Asymptotic theory is given for the MLE and for an information criterion that selects the number of groups and latent components, as both the number and the size of clusters grow.

2020

J43 Corrected Bayesian information criterion for stochastic block models [net] [#K]

J. Hu, H. Qin, T. Yan, Y. Zhao (2020). *J. Amer. Statist. Assoc. 115(532), 1771–1783*

A corrected BIC for SBMs, with a penalty adjusted for the number of blocks, that provably selects the number of communities consistently.

J44 Bayesian repulsive Gaussian mixture model [#K]

F. Xie, Y. Xu (2020). *J. Amer. Statist. Assoc. 115(529), 187–203*

Repulsive priors on component centres. Posterior consistency and contraction rates for density estimation are proved, and the repulsion shrinks the tail of the posterior on the number of components more than independent priors do.

J45 Hierarchical normalized completely random measures to cluster grouped data [#K]

R. Argiento, A. Cremaschi, M. Vannucci (2020). *J. Amer. Statist. Assoc. 115(529), 318–333*

Hierarchical normalised completely random measures for clustering grouped data, with clusters shared across groups. The partially exchangeable partition probability function and the prior distribution of the number of clusters are derived, together with a marginal sampler.

Selected landmark papers before 2020

J46 Probabilistic community detection with unknown number of communities [net] [#K]

J. Geng, A. Bhattacharya, D. Pati (2019). *J. Amer. Statist. Assoc. 114(526), 893–905*

A mixture-of-finite-mixtures SBM that infers the number of communities and the partition together. Non-asymptotic Bayes-risk bounds on the space of configurations hold even when K is unknown, and an efficient MCMC sampler avoids reversible jumps.

J47 Network cross-validation for determining the number of communities in network data [net] [#K]

K. Chen, J. Lei (2018). *J. Amer. Statist. Assoc. 113(521), 241–251*

Node-pair splitting cross-validation (NCV) for selecting K in SBMs and DCSBMs, with consistency guarantees, in particular against underestimation.

J48 Mixture models with a prior on the number of components [#K]

J. W. Miller, M. T. Harrison (2018). *J. Amer. Statist. Assoc. 113(521), 340–356*

Mixtures of finite mixtures share in simple closed form the key properties of Dirichlet-process mixtures: the EPPF, the restaurant process and the random-measure and stick-breaking representations. This lets DPM-style inference be used for the number of components.

J49 Inference on the order of a normal mixture [#K]

J. Chen, P. Li, Y. Fu (2012). *J. Amer. Statist. Assoc. 107(499), 1096–1105*

An EM test for the order of a normal mixture with unequal variances, with a tractable limiting distribution.

J50 Testing the order of a finite mixture [#K]

P. Li, J. Chen (2010). *J. Amer. Statist. Assoc. 105(491), 1084–1092*

An EM test of the null that the mixture has order m_0 against larger orders, with a tractable chi-square-mixture limit.

J51 Statistical significance of clustering for high-dimension, low-sample size data [#K]

Y. Liu, D. N. Hayes, A. Nobel, J. S. Marron (2008). *J. Amer. Statist. Assoc. 103(483), 1281–1293*

SigClust tests one Gaussian cluster against more, by calibrating the 2-means cluster index under a Gaussian null whose covariance is estimated in high-dimension, low-sample-size settings.

J52 Order selection in finite mixture models with a nonsmooth penalty [#K]

J. Chen, A. Khalili (2008). *J. Amer. Statist. Assoc. 103(484), 1674–1683*

Penalised likelihood with a nonsmooth penalty on the differences between ordered component parameters. It merges redundant components and estimates the order consistently.

J53 Finding the number of clusters in a dataset: an information-theoretic approach [#K]

C. A. Sugar, G. M. James (2003). *J. Amer. Statist. Assoc. 98(463), 750–763*

The "jump" method: transformed distortion curves from rate-distortion theory jump at the true number of clusters, with theoretical justification for Gaussian mixtures.

J54 Consistency of single linkage for high-density clusters

J. A. Hartigan (1981). *J. Amer. Statist. Assoc. 76(374), 388–394*

Single linkage is fractionally consistent for high-density clusters in one dimension, but not in two or more dimensions.

3 Journal of the Royal Statistical Society, Series B

Coverage note. All issues 2020–2026 and the advance articles listed on 7 October 2026.

2026

R1 Causal k-means clustering

K. Kim, J. Kim, E. H. Kennedy (2026). *J. R. Stat. Soc. Ser. B, advance article, published online 27 April 2026*

Applies k-means to unknown counterfactual regression functions to find subgroups with similar treatment responses. A plug-in estimator has explicit convergence rates, and a bias-corrected estimator based on nonparametric efficiency theory (double machine learning) is $\sqrt{n}$-consistent and asymptotically normal.

R2 Modelling with categorical features via exact fusion and sparsity regularization [group]

K. Behdin, R. Benbaki, P. Radchenko, R. Mazumder (2026). *J. R. Stat. Soc. Ser. B, advance article, published online 9 April 2026*

ClusterLearn-L0 fuses the levels of high-cardinality categorical predictors (clustering their coefficients) and selects variables, using ℓ_0-type penalties solved by mixed-integer optimisation. The paper proves prediction-error bounds, improved rates under groupwise selection, and exact recovery of the true clustering of levels under a minimum-separation condition.

R3 Optimal clustering by Lloyd's algorithm for low-rank mixture model

Z. Lyu, D. Xia (2026). *J. R. Stat. Soc. Ser. B 88(1), 43–65*

For matrix-valued observations with low-rank cluster centres, Lloyd's algorithm combined with low-rank approximation, and initialised by a tensor-based spectral method, attains the minimax-optimal exponential clustering error, which is governed by the separation strength. There is evidence that no polynomial-time method is consistent when the signal (the smallest singular value of the centres) is too weak, even with strong separation.

2025

R4 Product centred Dirichlet processes for Bayesian multiview clustering

A. Dombowsky, D. B. Dunson (2025). *J. R. Stat. Soc. Ser. B 87(5), 1331–1352*

CLIC is a prior that models the dependence between the clusterings of different views with a single parameter, through a product centred Dirichlet process lying between independent and identical partitions. It has closed-form marginal and joint partition distributions, a finite approximation with proven accuracy, and a marginal Gibbs sampler.

R5 Bayesian mixture models with repulsive and attractive atoms

M. Beraha, R. Argiento, F. Camerlenghi, A. Guglielmi (2025). *J. R. Stat. Soc. Ser. B 87(5), 1481–1507*

A unified theory of random probability measures whose atoms interact, repulsively or attractively, through a general finite point process. Palm calculus gives closed-form posterior, marginal and predictive distributions, which are worked out for Poisson, Gibbs, determinantal and shot-noise Cox processes.

R6 Repulsion, chaos, and equilibrium in mixture models

A. Cremaschi, T. M. Wertz, M. De Iorio (2025). *J. R. Stat. Soc. Ser. B 87(2), 389–432*

Repulsive priors for component locations built from Gibbs measures, including joint eigenvalue laws of random matrices, with closed-form normalising constants. Their theoretical properties are studied through tools from statistical mechanics (large deviations and equilibrium/DLR-type characterisations).

2024

R7 Root and community inference on the latent growth process of a network [net]

H. Crane, M. Xu (2024). *J. R. Stat. Soc. Ser. B 86(4), 825–865*

(Discussion paper.) In the PAPER model, a preferential-attachment tree plus Erdős–Rényi noise edges, confidence sets are built for the root and, with several simultaneous growth processes, for the communities. The expected size of the confidence set stays small unless the noise is large.

R8 Full-model estimation for non-parametric multivariate finite mixture models [#K]
M. Du Roy de Chaumaray, M. Marbac (2024). *J. R. Stat. Soc. Ser. B 86(4), 896–921*
Discretises each variable and penalises the resulting log-likelihood to select both the number of components and the subset of discriminative variables in nonparametric mixtures with conditionally independent coordinates. The selected model is consistent for a suitable penalty and number of bins.

R9 Monte Carlo goodness-of-fit tests for degree corrected and related stochastic blockmodels [net]
V. Karwa, D. Pati, S. Petrović, L. Solus, N. Alexeev, M. Raič, D. Wilburne, R. Williams, B. Yan (2024). *J. R. Stat. Soc. Ser. B 86(1), 90–121*
Exact finite-sample goodness-of-fit tests for the Erdős–Rényi, additive and degree-corrected SBMs. Conditional p-values are obtained by sampling the fibres of the sufficient statistics with Markov bases from algebraic statistics, and are averaged over the posterior when blocks are latent; the number of blocks is taken as known.

2023

R10 Informative core identification in complex networks [net]
R. Miao, T. Li (2023). *J. R. Stat. Soc. Ser. B 85(1), 108–126*
A core–periphery model in which only an informative core carries structure such as communities. Spectral algorithms extract the core before community detection, with theoretical accuracy guarantees.

2022

R11 Exact clustering in tensor block model: statistical optimality and computational limit
R. Han, Y. Luo, M. Wang, A. R. Zhang (2022). *J. R. Stat. Soc. Ser. B 84(5), 1666–1698*
A high-order Lloyd algorithm with high-order spectral initialisation achieves exact clustering in tensor block models at the optimal signal level, among computationally feasible methods. Matching computational lower bounds show a statistical–computational gap.

R12 A statistical interpretation of spectral embedding: the generalised random dot product graph [net]
P. Rubin-Delanchy, J. Cape, M. Tang, C. E. Priebe (2022). *J. R. Stat. Soc. Ser. B 84(4), 1446–1473*
Under the GRDPG, which allows heterophily through indefinite inner products, adjacency and Laplacian spectral embeddings estimate latent positions uniformly and satisfy a central limit theorem. Communities therefore appear as Gaussian clusters, and mixed memberships fill a simplex, which justifies Gaussian-mixture clustering of the embedding.

R13 Semiparametric latent class analysis of recurrent event data
W. Zhao, L. Peng, J. Hanfelt (2022). *J. R. Stat. Soc. Ser. B 84(4), 1175–1197*
Latent classes with class-specific semiparametric models for recurrent-event rates, estimated robustly with a conditional-score technique. Asymptotic theory is established for the resulting estimators.

R14 Supervised multivariate learning with simultaneous feature auto-grouping and dimension reduction [group] [#K]
Y. She, J. Shen, C. Zhang (2022). *J. R. Stat. Soc. Ser. B 84(3), 912–932*
Clustered reduced-rank learning groups features automatically while building predictive factors. The paper derives information-theoretic limits that show the intrinsic cost of seeking clusters, non-asymptotic accuracy guarantees for fixed-point estimators, and an information criterion with non-asymptotic theory for choosing the number of clusters and the rank.

R15 SIMPLE: statistical inference on membership profiles in large networks [net]
J. Fan, Y. Fan, X. Han, J. Lv (2022). *J. R. Stat. Soc. Ser. B 84(2), 630–653*
Tests whether two given nodes share the same membership profile under degree-corrected mixed-membership models. Spectral test statistics have explicit (chi-square-type) null limits, which gives valid inference on community membership.

2021

R16 Modelling high-dimensional categorical data using nonconvex fusion penalties [group]
B. G. Stokell, R. D. Shah, R. J. Tibshirani (2021). *J. R. Stat. Soc. Ser. B 83(3), 579–611*

SCOPE clusters the levels of each categorical predictor through a nonconvex (minimax concave) fusion penalty, which is solved exactly by dynamic programming in the univariate case. Oracle-type theory shows that the true grouping of levels is recovered in high dimensions.

R17 Estimation and clustering in popularity adjusted block model [net]

M. Noroozi, R. Rimal, M. Pensky (2021). *J. R. Stat. Soc. Ser. B 83(2), 293–317*

The popularity adjusted block model, which generalises the DCSBM, is linked to subspace clustering. The paper gives estimation-error and clustering-consistency guarantees for its community detection.

2020

R18 Exchangeable random measures for sparse and modular graphs with overlapping communities [net]

A. Todeschini, X. Miscouridou, F. Caron (2020). *J. R. Stat. Soc. Ser. B 82(2), 487–520*

Nodes carry community-affiliation weights in a model built on exchangeable random measures. The paper proves sparsity and power-law degree properties for graphs with overlapping communities, and develops posterior inference.

R19 Targeted sampling from massive block model graphs with personalized PageRank [net]

F. Chen, Y. Zhang, K. Rohe (2020). *J. R. Stat. Soc. Ser. B 82(1), 99–126*

Under the DCSBM, personalised PageRank from a seed node is biased toward high-degree nodes outside the seed's block. A degree adjustment corrects this, and consistency of PageRank-based clustering, for directed graphs as well, is proved via element-wise eigenvector bounds.

Selected landmark papers before 2020

R20 On choosing mixture components via non-local priors [#K]

J. Fúquene, M. Steel, D. Rossell (2019). *J. R. Stat. Soc. Ser. B 81(5), 809–837*

Non-local priors for mixtures favour well-separated components with non-negligible weight. Bayes factors for the number of components are shown to be ratios of posterior to prior empty-cluster probabilities, and the sparsity these priors induce is characterised theoretically.

R21 Convex clustering via ℓ_1 fusion penalization [#K]

P. Radchenko, G. Mukherjee (2017). *J. R. Stat. Soc. Ser. B 79(5), 1527–1546*

Represents convex (ℓ_1-fusion) clustering as a sequence of splits and proves that it consistently estimates its population analogue, with rates. A post-processing step provably estimates the number of clusters.

R22 A Bayesian information criterion for singular models [#K]

M. Drton, M. Plummer (2017). *J. R. Stat. Soc. Ser. B 79(2), 323–380*

(Discussion paper.) sBIC uses learning coefficients (real log-canonical thresholds) to approximate the marginal likelihood of singular models, such as mixtures with surplus components. It is consistent for selecting the number of mixture components.

R23 Hypothesis testing for automated community detection in networks [net] [#K]

P. J. Bickel, P. Sarkar (2016). *J. R. Stat. Soc. Ser. B 78(1), 253–273*

The largest eigenvalue of a centred and scaled adjacency matrix has a Tracy–Widom limit under the Erdős–Rényi null, which gives a test of $K = 1$. Applied recursively, the test bipartitions the network and thereby determines the number of communities.

R24 Non-parametric identification and estimation of the number of components in multivariate mixtures [#K]

H. Kasahara, K. Shimotsu (2014). *J. R. Stat. Soc. Ser. B 76(1), 97–111*

With conditionally independent coordinates, the rank of certain matrices of marginal distributions gives a lower bound on the number of components, and sequential rank tests estimate that bound consistently.

R25 Asymptotic behaviour of the posterior distribution in overfitted mixture models [#K]

J. Rousseau, K. Mengersen (2011). *J. R. Stat. Soc. Ser. B 73(5), 689–710*

When the mixture is overfitted and the Dirichlet weight hyperparameter is small enough (below $d/2$), the posterior empties the superfluous components, so the effective number of components is learned.

R26 A modified likelihood ratio test for homogeneity in finite mixture models [#K]

H. Chen, J. Chen, J. D. Kalbfleisch (2001). *J. R. Stat. Soc. Ser. B 63(1), 19–29*

A penalised (modified) likelihood ratio test of one component against two, with a simple chi-square-mixture limit.

R27 Estimating the number of clusters in a data set via the gap statistic [#K]

R. Tibshirani, G. Walther, T. Hastie (2001). *J. R. Stat. Soc. Ser. B 63(2), 411–423*

The gap statistic compares the log within-cluster dispersion with its expectation under a reference null distribution, and theoretical arguments support the uniform reference.

R28 Using kernel density estimates to investigate multimodality [#K]

B. W. Silverman (1981). *J. R. Stat. Soc. Ser. B 43(1), 97–99*

The critical bandwidth (the smallest bandwidth giving at most k modes) is used to test the number of modes, with bootstrap calibration.

4 Biometrika

Coverage note. All issues 2020–2026 and the advance articles listed on 6–7 October 2026. Recent articles that have no final pagination yet are cited by their article identifier (e.g. asag018).

2026

B1 Dynamic clustering for heterophilic stochastic block models with time-varying node memberships [net]

K. Z. Lin, J. Lei (2026). *Biometrika 113(2), asag018*

Sequences of SBMs in which nodes gradually switch communities and each snapshot alone carries little signal. KD-SoS, a kernel-weighted, debiased sum-of-squares spectral method, detects the communities consistently in every network, without assuming smooth changes in heterophilic connectivity. The paper also shows how communities can be aligned across time according to the switching rates.

B2 Testing for latent structure via the Wilcoxon–Wigner random matrix of normalized rank statistics [net] [#K]

J. Z. Liao, J. Cape (2026). *Biometrika 113(2), asag003*

Wilcoxon–Wigner matrices are symmetric matrices of normalised rank statistics from an i.i.d. sample. Their leading eigenvalue and eigenvector have Gaussian fluctuations with explicit centring and scaling, which give distribution-free spectral tests for community structure (none against some) and for principal-submatrix detection.

2025

B3 Bayesian clustering of high-dimensional data via latent repulsive mixtures

L. Ghilotti, M. Beraha, A. Guglielmi (2025). *Biometrika 112(2), 551–558*

Clusters are modelled through a mixture on latent factor scores, with a repulsive, anisotropic determinantal point-process prior on the cluster centres. The paper constructs anisotropic DPPs with tractable densities and shows that repulsion makes the clustering more robust to misspecification than standard latent-factor mixtures.

2024

B4 Network-adjusted covariates for community detection [net]

Y. Hu, W. Wang (2024). *Biometrika 111(4), 1221–1240*

Network-adjusted covariates combine each node's covariates with its connections, using tuning-free node-specific weights. Strong consistency holds under DCSBMs with covariates, even with misspecification and several sparse communities, and a general lower bound shows optimality in connectivity up to a constant.

B5 Network community detection using higher-order structures [net]

X. Yu, J. Zhu (2024). *Biometrika 111(3), 903–923*

An edge-dependent network model with both community and triangle structure. Triangle-based community detection has a finite-sample error bound governed by the expected triangle degree, and it is consistent; these are the first such results for a single network with dependent edges.

2023

B6 A generalized Bayes framework for probabilistic clustering

T. Rigon, A. H. Herring, D. B. Dunson (2023). *Biometrika 110(3), 559–578*

Gibbs posteriors replace the log-likelihood with a clustering loss, based on Bregman divergences or pairwise similarities, so that k-means-type algorithms become generalised Bayes estimators. This gives coherent uncertainty quantification, such as the probability that a point is well clustered.

B7 Clustering consistency with Dirichlet process mixtures [#K]

F. Ascolani, A. Lijoi, G. Rebaudo, G. Zanella (2023). *Biometrika 110(2), 551–558*

Dirichlet process mixtures with a fixed concentration parameter overestimate the number of clusters.

With a prior on the concentration parameter, the posterior on the number of clusters is consistent for the true number of components in well-specified finite mixtures, under mild conditions.

B8 Spherical clustering in detection of groups of concomitant extremes
V. Fomichov, J. Ivanovs (2023). *Biometrika 110(1), 135–153*
Spherical k-means applied to the angular parts of large observations identifies groups of variables that are extreme together (the support directions of the spectral measure). The procedure comes with consistency results.

2022

No qualifying article identified.

2021

B9 Learning block structures in U-statistic-based matrices [#K]
W. Zhang, B. Jin, Z. Bai (2021). *Biometrika 108(4), 933–946*
Groups of variables form block structures in U-statistic-based association matrices, such as Kendall's tau. The number of blocks is estimated and the blocks are recovered, with consistency guarantees in high dimensions.

B10 The asymptotic distribution of modularity in weighted signed networks [net] [#K]
R. Ma, I. Barnett (2021). *Biometrika 108(1), 1–16*
Modularity is asymptotically normal for weighted, signed networks under the null of no community structure, which gives a test of whether communities exist ($K = 1$ against $K > 1$).

2020

B11 The Pitman–Yor multinomial process for mixture modelling
A. Lijoi, I. Prünster, T. Rigon (2020). *Biometrika 107(4), 891–906*
A finite-dimensional analogue of the Pitman–Yor process is proposed as a mixing prior. Its partition structure, including the distribution of the number of clusters, is derived, as is its convergence to the Pitman–Yor process.

B12 Network cross-validation by edge sampling [net] [#K]
T. Li, E. Levina, J. Zhu (2020). *Biometrika 107(2), 257–276*
(Discussion paper.) Edge cross-validation holds out random node pairs, completes the matrix with a low-rank method and evaluates the held-out entries. It provably selects the number of communities consistently and chooses correctly between models such as SBM and DCSBM.

B13 Consistent community detection in multi-layer network data [net]
J. Lei, K. Chen, B. Lynch (2020). *Biometrika 107(1), 61–73*
Multilayer SBMs with common communities. The least-squares estimator is consistent even when each layer is too sparse to be informative alone, provided the number of layers grows.

Selected landmark papers before 2020

B14 Covariate-assisted spectral clustering [net]
N. Binkiewicz, J. T. Vogelstein, K. Rohe (2017). *Biometrika 104(2), 361–377*
Spectral clustering of a combination of the regularised graph Laplacian and a node-covariate Gram matrix, with misclustering bounds under a joint SBM–covariate model.

B15 Stochastic blockmodels with a growing number of classes [net]
D. S. Choi, P. J. Wolfe, E. M. Airoldi (2012). *Biometrika 99(2), 273–284*
Likelihood-based community detection in SBMs is consistent even when the number of classes grows with the number of nodes.

B16 Consistent selection of the number of clusters via crossvalidation [#K]
J. Wang (2010). *Biometrika 97(4), 893–904*
Clustering instability, measured by cross-validation, selects the number of clusters consistently.

B17 Testing the number of components in a normal mixture [#K]

Y. Lo, N. R. Mendell, D. B. Rubin (2001). *Biometrika 88(3), 767–778*

A likelihood-ratio test of k_0 against k_1 components, based on the asymptotic distribution of the statistic under misspecification (a weighted sum of chi-squares).

5 Journal of Machine Learning Research

Coverage note. Volumes 21 (2020) to 27 (2026, papers published by early October 2026). JMLR publishes continuously, so within a year articles are listed by decreasing paper number.

2026

M1 Bayesian level set clustering

D. Buch, M. Dewaskar, D. B. Dunson (2026). *J. Mach. Learn. Res. 27(159)*

Clusters are defined as connected components of a level set of the data density, which separates clustering from flexible Bayesian density modelling: mixture components are not clusters. Clustering consistency is established.

M2 Simultaneous identification of sparse structures and communities in heterogeneous graphical models

D. Shi, T. Wang, Z. Ying (2026). *J. Mach. Learn. Res. 27(148)*

Decomposes a Gaussian graphical model into low-rank diagonal blocks plus a sparse part, to find non-overlapping communities of variables. The paper gives local identifiability, an adaptive irrepresentability condition for selection consistency, and clustering-error bounds for its k-means stage.

M3 Graph-based clustering revisited: a relaxation of kernel k-means perspective

W. Lyu, Y. Jia, H. Liu, J. Hou (2026). *J. Mach. Learn. Res. 27(131)*

Recasts graph-based clustering as a relaxation of kernel k-means that relaxes only the orthonormality constraint (LoRD, B-LoRD). The paper proves that orthogonality and block-diagonality are equivalent under doubly stochastic constraints, and gives convergence guarantees for the projected-gradient algorithm.

M4 Multi-relational network autoregression model with latent group structures [#K]

Y. Ren, X. Zhu, G. Xu, Y. Ma (2026). *J. Mach. Learn. Res. 27(78)*

Nodes share autoregression parameters within latent groups that differ across relations. Memberships and group-wise parameters are consistent even when the number of groups is over-specified, and an information criterion selects the number of groups consistently.

M5 Two-way node popularity model for directed and bipartite networks [net]

B.-Y. Jing, T. Li, J. Wang, Y. Wang (2026). *J. Mach. Learn. Res. 27(43)*

Communities in directed and bipartite networks whose nodes have community-specific popularity, with edges from a general sub-Gaussian family. Theory for the proposed fitting methods (Delete-One-Method and a two-stage divided cosine algorithm) covers estimation and community-recovery accuracy.

2025

M6 Randomly projected convex clustering model: motivation, realization, and cluster recovery guarantees

Z. Wang, Y. Yuan, J. Ma, T. Zeng, D. Sun (2025). *J. Mach. Learn. Res. 26(137)*

Convex clustering after random projection keeps the perfect recovery of the original space. Exact recovery holds with embedding dimension $O(\epsilon^{-2} \log K)$, which does not depend on the number of data points.

M7 Bagged k-distance for mode-based clustering using the probability of localized level sets

H. Hang (2025). *J. Mach. Learn. Res. 26(97)*

An ensemble (bagged k-distance) mode-based clustering method that uses a single threshold across varying densities. It attains optimal rates for mode estimation and fast Hausdorff rates for localised level-set estimation.

M8 Determine the number of states in hidden Markov models via marginal likelihood [#K]

Y. Chen, C.-D. Fuh, C.-L. M. Kao (2025). *J. Mach. Learn. Res. 26(58)*

Selects the order of an HMM by the marginal likelihood, integrating out both parameters and hidden

states. The selector is consistent, including for Gaussian HMMs with heterogeneous covariances, and it covers mixture-order selection as a special case.

M9 Curvature-based clustering on graphs [net]

Y. Tian, Z. Lubberts, M. Weber (2025). *J. Mach. Learn. Res. 26(52)*

Node clustering with discrete Ricci curvatures and geometric flows. Mixed-membership detection runs on the line graph, and is justified by a relation between the curvature of a graph and that of its dual.

M10 Bayesian sparse Gaussian mixture model for clustering in high dimensions [#K]

D. Yao, F. Xie, Y. Xu (2025). *J. Mach. Learn. Res. 26(21)*

Sparse high-dimensional Gaussian mixtures in which the number of clusters may grow and is estimated adaptively. The paper gives a minimax lower bound for parameter estimation, a spike-and-slab posterior that contracts at the minimax rate, and a misclustering-rate bound.

M11 An axiomatic definition of hierarchical clustering

E. Arias-Castro, E. Coda (2025). *J. Mach. Learn. Res. 26(10)*

An axiomatic definition of population hierarchical clustering for piecewise-constant densities, extended to general densities much as the Lebesgue integral is. Under regularity conditions it recovers Hartigan's cluster tree.

2024

M12 Estimation of the order of non-parametric hidden Markov models using the singular values of an integral operator [#K]

M. Du Roy de Chaumaray, S. El Kolei, M.-P. Etienne, M. Marbac (2024). *J. Mach. Learn. Res. 25(415)*

The number of hidden states equals the rank of an integral operator. A data-driven thresholding of its estimated singular values estimates the order consistently, with no a-priori upper bound required.

M13 Graphical Dirichlet process for clustering non-exchangeable grouped data

A. Chakrabarti, Y. Ni, E. R. A. Morris, M. L. Salinas, R. S. Chapkin, B. K. Mallick (2024). *J. Mach. Learn. Res. 25(323)*

Dirichlet processes for groups linked by a known DAG, with each group's concentration and base measure depending on its parents. The paper derives hypergraph, stick-breaking and restaurant-type representations, a finite-mixture limit, and a Markov property with respect to the DAG.

M14 Contamination-source based K-sample clustering

X. Milhaud, D. Pommeret, Y. Salhi, P. Vandekerkhove (2024). *J. Mach. Learn. Res. 25(287)*

A semiparametric method that clusters K populations according to the unknown contamination component they share, with consistency of the clustering when true clusters exist.

M15 Heterogeneity-aware clustered distributed learning for multi-source data analysis [group] [#K]

Y. Chen, Q. Zhang, S. Ma, K. Fang (2024). *J. Mach. Learn. Res. 25(211)*

Clients (data sources) are clustered into groups that share a model, using group and fusion penalties inside a distributed algorithm. The number of clusters is not prespecified, and estimation, selection and clustering are all proved consistent.

M16 Optimal clustering with bandit feedback

J. Yang, Z. Zhong, V. Y. F. Tan (2024). *J. Mach. Learn. Res. 25(186)*

Arms with unknown cluster-specific means are sampled sequentially in order to cluster them. The paper gives an instance-dependent lower bound on the sample complexity and an asymptotically optimal algorithm.

M17 Fermat distances: metric approximation, spectral convergence, and clustering algorithms

N. García Trillos, A. Little, D. McKenzie, J. M. Murphy (2024). *J. Mach. Learn. Res. 25(176)*

Power-weighted shortest-path (Fermat) distances converge to a density-sensitive continuum metric, and the associated graph Laplacians converge spectrally. This gives guarantees for Fermat-distance clustering.

M18 Variational estimators of the degree-corrected latent block model for bipartite networks [net]

Y. Zhao, N. Hao, J. Zhu (2024). *J. Mach. Learn. Res. 25(150)*

Variational estimators for co-clustering bipartite networks under a degree-corrected latent block model, with consistency of the row and column labels and asymptotic properties of the parameter estimates.

M19 Sum-of-norms clustering does not separate nearby balls

A. Dunlap, J.-C. Mourrat (2024). *J. Mach. Learn. Res. 25(123)*

A negative result: a continuum version of sum-of-norms (convex) clustering cannot separate two nearby uniform balls, even disjoint ones, which limits the method's resolution.

M20 Pursuit of the cluster structure of network lasso: recovery condition and non-convex extension

S. Yagishita, J. Gotoh (2024). *J. Mach. Learn. Res. 25(21)*

Sufficient conditions for network lasso to recover the true cluster structure exactly, and a nonconvex extension that recovers it under weaker conditions.

2023

M21 Distributed community detection in large networks [net]

S. Zhang, R. Song, W. Lu, J. Zhu (2023). *J. Mach. Learn. Res. 24(401)*

A divide-and-conquer algorithm that provably recovers both the group and the community structure of networks with hierarchical organisation, at a fraction of the computational cost.

M22 Large data limit of the MBO scheme for data clustering: convergence of the dynamics

T. Laux, J. Lelmi (2023). *J. Mach. Learn. Res. 24(344)*

As the number of data points grows, the dynamics of the graph MBO clustering scheme converge to a viscosity solution of (multiphase) mean curvature flow.

M23 Community recovery in the geometric block model [net]

S. Galhotra, A. Mazumdar, S. Pal, B. Saha (2023). *J. Mach. Learn. Res. 24(338)*

In the geometric block model, simple triangle-counting recovers communities near-optimally, especially in the logarithmic-degree regime, where it beats SBM-designed methods.

M24 Higher-order spectral clustering under superimposed stochastic block models [net]

S. Paul, O. Milenkovic, Y. Chen (2023). *J. Mach. Learn. Res. 24(320)*

Networks with superimposed edges and higher-order structures, such as triangles and hyperedges. Spectral clustering on higher-order (motif) adjacency matrices has consistency and misclustering guarantees that show when the higher-order information helps.

M25 Community models for networks observed through edge nominations [net]

T. Li, E. Levina, J. Zhu (2023). *J. Mach. Learn. Res. 24(282)*

A nomination SBM for networks sampled by nominations. Spectral clustering, together with method-of-moments fitting, recovers the communities consistently.

M26 Adaptive clustering using kernel density estimators [#K]

I. Steinwart, B. K. Sriperumbudur, P. Thomann (2023). *J. Mach. Learn. Res. 24(275)*

A generic recursive algorithm that estimates all the splits of a finite cluster tree from kernel density estimates, consistently and with adaptive rates and no continuity assumptions. The clusters at each split, and hence their number, are determined by the data.

M27 Clustering with tangles: algorithmic framework and theoretical guarantees

S. Klepper, C. Elbracht, D. Fioravanti, J. Kneip, L. Rendsburg, M. Teegen, U. von Luxburg (2023). *J. Mach. Learn. Res. 24(190)*

Adapts tangles from graph-minor theory into a clustering framework that aggregates many weak bipartitions. Recovery guarantees are proved under SBM- and mixture-type models.

M28 Consistent model-based clustering using the quasi-Bernoulli stick-breaking process [#K]

C. Zeng, J. W. Miller, L. L. Duan (2023). *J. Mach. Learn. Res. 24(153)*

A stick-breaking prior under which the posterior on the number of clusters is consistent, unlike Dirichlet- and Pitman–Yor-process mixtures, with no extra computational cost.

M29 Selective inference for k-means clustering

Y. T. Chen, D. M. Witten (2023). *J. Mach. Learn. Res. 24(152)*

A selective p-value for the difference in means between two k-means clusters, conditioning on the clustering. It controls the selective type I error exactly and is computed efficiently.

M30 Escaping the curse of dimensionality in Bayesian model-based clustering

N. K. Chandra, A. Canale, D. B. Dunson (2023). *J. Mach. Learn. Res. 24(144)*

As the dimension grows, posteriors of standard Bayesian mixtures degenerate: all points end up in one cluster or in singletons. Clustering on low-dimensional latent factors (Lamb) avoids this, with theory for the induced partition.

M31 Large sample spectral analysis of graph-based multi-manifold clustering

N. García Trillos, P. He, C. Li (2023). *J. Mach. Learn. Res. 24(143)*

Graph Laplacians built with tangent-aware (local PCA) weights on data from intersecting manifolds converge spectrally to a limit that separates the manifolds, which justifies multi-manifold spectral clustering.

M32 The hyperspherical geometry of community detection: modularity as a distance [net]

M. Gösgens, R. van der Hofstad, N. Litvak (2023). *J. Mach. Learn. Res. 24(112)*

Clusterings are embedded as vectors on a hypersphere, and modularity maximisation is shown to be equivalent to minimising the angular distance to a modularity vector. This gives a geometric view of community detection.

M33 Contextual stochastic block model: sharp thresholds and contiguity [net]

C. Lu, S. Sen (2023). *J. Mach. Learn. Res. 24(54)*

Sharp information-theoretic thresholds for detecting communities in a sparse SBM observed together with high-dimensional Gaussian node covariates, proved by contiguity arguments.

M34 A group-theoretic approach to computational abstraction: symmetry-driven hierarchical clustering

H. Yu, I. Mineyev, L. R. Varshney (2023). *J. Mach. Learn. Res. 24(47)*

Group actions (symmetries) induce hierarchies of partitions, i.e. abstractions, which gives a lattice- and group-theoretic framework for hierarchical clustering.

M35 Bayesian spiked Laplacian graphs [net]

L. L. Duan, G. Michailidis, M. Ding (2023). *J. Mach. Learn. Res. 24(3)*

A network model whose graph Laplacian has a spiked spectrum, so that community detection with uncertainty quantification becomes posterior inference on the leading eigenvectors. Posterior consistency is established.

M36 Approximation bounds for hierarchical clustering: average linkage, bisecting k-means, and local search

B. Moseley, J. R. Wang (2023). *J. Mach. Learn. Res. 24(1)*

Approximation guarantees, and their limits, for average linkage, bisecting k-means and local search under a revenue objective for hierarchical clustering that is dual to Dasgupta's cost.

2022

M37 ALMA: alternating minimization algorithm for clustering mixture multilayer network [net]

X. Fan, M. Pensky, F. Yu, T. Zhang (2022). *J. Mach. Learn. Res. 23(330)*

Mixture multilayer SBMs. Alternating minimisation recovers the partition of the layers and estimates the connection-probability matrices, with error bounds that improve on tensor-based TWIST.

M38 Community detection in sparse latent space models [net]

F. Gao, Z. Ma, H. Yuan (2022). *J. Mach. Learn. Res. 23(322)*

Spectral clustering followed by refinement through normalised edge counts is consistent, and even optimal, for a broad class of sparse latent space models. The proof uses an equivalence between likelihood-ratio testing and edge counting.

M39 Detecting latent communities in network formation models [net]

S. Ma, L. Su, Y. Zhang (2022). *J. Mach. Learn. Res. 23(310)*

A logistic network-formation model with assortative matching, latent community structure in the coefficients, and low-rank edge effects. The latent communities are recovered exactly once the expected degree is of order $\log n$ or larger.

M40 Theoretical foundations of t-SNE for visualizing high-dimensional clustered data
T. T. Cai, R. Ma (2022). *J. Mach. Learn. Res. 23(301)*
The early-exaggeration stage of t-SNE acts as power iteration on a graph Laplacian, i.e. implicit spectral clustering. The later stage has an amplification phase that separates clusters, followed by stabilisation, and the paper gives conditions under which the visual clusters are faithful.

M41 Exact partitioning of high-order models with a novel convex tensor cone relaxation [net]
C. Ke, J. Honorio (2022). *J. Mach. Learn. Res. 23(284)*
A convex relaxation over a tensor cone exactly recovers planted partitions in high-order (hypergraph-type) models with high probability, under explicit conditions.

M42 Clustering with semidefinite programming and fixed point iteration
P. Felzenszwalb, C. Klivans, A. Paul (2022). *J. Mach. Learn. Res. 23(190)*
In the SDP relaxation of Max k-Cut, the vertices correspond to partitions into k sets and are attractive fixed points of an iterated linear-optimisation rounding scheme.

M43 Regularized k-means through hard-thresholding
J. Raymaekers, R. H. Zamar (2022). *J. Mach. Learn. Res. 23(93)*
k-means that directly penalises the size of the cluster centres. Different penalties are compared theoretically, and the comparison leads to hard-threshold k-means, which uses an ℓ_0 penalty to select variables.

M44 Dependent randomized rounding for clustering and partition systems with knapsack constraints
D. G. Harris, T. Pensyl, A. Srinivasan, K. Trinh (2022). *J. Mach. Learn. Res. 23(81)*
Dependent randomised rounding of LP relaxations gives approximation algorithms, with proven ratios, for k-center- and k-median-type clustering under knapsack and partition constraints.

2021

M45 Regularized spectral methods for clustering signed networks [net]
M. Cucuringu, A. V. Singh, D. Sulem, H. Tyagi (2021). *J. Mach. Learn. Res. 22(264)*
Extends spectral methods for signed networks to k-way clustering with unequal sizes, with regularised variants for sparse graphs. Recovery guarantees hold under a signed SBM.

M46 Statistical guarantees for local spectral clustering on random neighborhood graphs
A. Green, S. Balakrishnan, R. J. Tibshirani (2021). *J. Mach. Learn. Res. 22(247)*
Personalised PageRank on neighbourhood graphs of i.i.d. samples recovers density clusters that are well-conditioned at the population level (small normalised cut, large conductance and local spread).

M47 Learning with semi-definite programming: statistical bounds based on fixed point analysis and excess risk curvature [net]
S. Chrétien, M. Cucuringu, G. Lecué, L. Neirac (2021). *J. Mach. Learn. Res. 22(230)*
General estimation and prediction bounds for SDP estimators, obtained from fixed-point and excess-risk- curvature arguments. Applications include community detection in SBMs, signed clustering and MAX-CUT.

M48 A unified framework for spectral clustering in sparse graphs [net] [#K]
L. Dall'Amico, R. Couillet, N. Tremblay (2021). *J. Mach. Learn. Res. 22(217)*
A suitably parametrised regularised Laplacian, related to the Bethe Hessian, clusters sparse and degree-heterogeneous networks. It also estimates the number of communities as well as reconstructing them.

M49 Sparse popularity adjusted stochastic block model [net]
M. Noroozi, M. Pensky, R. Rimal (2021). *J. Mach. Learn. Res. 22(193)*
A sparse version of the popularity adjusted block model, in which some community pairs are

disconnected, with clustering via sparse subspace clustering and error and consistency guarantees.

M50 Locally private k-means clustering
U. Stemmer (2021). *J. Mach. Learn. Res. 22(176)*
k-means under local differential privacy with improved additive error, and lower bounds that show the error is nearly optimal.

M51 Kernel smoothing, mean shift, and their learning theory with directional data
Y. Zhang, Y.-C. Chen (2021). *J. Mach. Learn. Res. 22(154)*
Theory for directional kernel density estimation and its derivatives on the sphere, with convergence of the directional mean-shift algorithm and consistency of mode-based clustering on the sphere.

M52 Statistical guarantees for local graph clustering [net]
W. Ha, K. Fountoulakis, M. W. Mahoney (2021). *J. Mach. Learn. Res. 22(148)*
Recovery guarantees for ℓ_1-regularised PageRank local clustering of a target cluster under random graph models.

M53 On efficient multilevel clustering via Wasserstein distances
V. Huynh, N. Ho, N. Dam, X. Nguyen, M. Yurochkin, H. Bui, D. Phung (2021). *J. Mach. Learn. Res. 22(145)*
Multilevel clustering, of the data within groups and of the groups themselves, through Wasserstein barycentres, with consistency results for the estimators.

M54 Strong consistency, graph Laplacians, and the stochastic block model [net]
S. Deng, S. Ling, T. Strohmer (2021). *J. Mach. Learn. Res. 22(117)*
Spectral clustering with unnormalised and normalised graph Laplacians achieves exact recovery (strong consistency) near the information-theoretic threshold, via ℓ_∞ eigenvector perturbation.

M55 Geometric structure of graph Laplacian embeddings
N. García Trillos, F. Hoffmann, B. Hosseini (2021). *J. Mach. Learn. Res. 22(63)*
For well-separated clusters, graph Laplacian embeddings concentrate near orthogonal cones (rays) in a quantified way, which explains the success of spectral clustering.

M56 Subspace clustering through sub-clusters
W. Li, J. Hannig, S. Mukherjee (2021). *J. Mach. Learn. Res. 22(53)*
A sampling-based approach clusters a random subsample with a base subspace-clustering method and then assigns the remaining points. It comes with guarantees for exact clustering at much lower computational cost.

M57 A Bayesian contiguous partitioning method for learning clustered latent variables
Z. T. Luo, H. Sang, B. Mallick (2021). *J. Mach. Learn. Res. 22(37)*
Spatially contiguous partitions are built from random spanning trees, and posterior concentration is established for the clustered latent variables.

M58 When random initializations help: a study of variational inference for community detection [net]
P. Sarkar, Y. X. R. Wang, S. S. Mukherjee (2021). *J. Mach. Learn. Res. 22(22)*
In two-block SBMs, mean-field variational inference from a random initialisation can converge to the truth or fail. Many random initialisations together succeed with high probability.

M59 Homogeneity structure learning in large-scale panel data with heavy-tailed errors [group]
D. Xiao, Y. Ke, R. Li (2021). *J. Mach. Learn. Res. 22(13)*
Interactive-effects panel models with endogeneity and heavy tails. A robust procedure learns a homogeneity (grouping) structure of high-dimensional individual covariate effects, with non-asymptotic guarantees.

M60 Mixing time of Metropolis–Hastings for Bayesian community detection [net]
B. Zhuo, C. Gao (2021). *J. Mach. Learn. Res. 22(10)*
Under signal conditions, a Metropolis–Hastings sampler for the SBM posterior mixes rapidly, so Bayesian community detection is computationally feasible.

M61 Convex clustering: model, theoretical guarantee and efficient algorithm

D. Sun, K.-C. Toh, Y. Yuan (2021). *J. Mach. Learn. Res. 22(9)*

Deterministic conditions under which convex clustering recovers the clusters exactly, and a semismooth Newton augmented Lagrangian algorithm to compute it.

2020

M62 Recovery of a mixture of Gaussians by sum-of-norms clustering

T. Jiang, S. Vavasis, C. W. Zhai (2020). *J. Mach. Learn. Res. 21(225)*

Sum-of-norms (convex) clustering recovers the components of a Gaussian mixture with high probability when the means are sufficiently separated relative to the noise.

M63 Provable convex co-clustering of tensors

E. C. Chi, B. J. Gaines, W. W. Sun, H. Zhou, J. Yang (2020). *J. Mach. Learn. Res. 21(214)*

CoCo, a convex co-clustering estimator for tensors, with finite-sample error bounds and consistent co-cluster recovery.

M64 Spectral algorithms for community detection in directed networks [net]

Z. Wang, Y. Liang, P. Ji (2020). *J. Mach. Learn. Res. 21(153)*

Spectral methods based on ratios of singular vectors for degree-corrected directed networks, with consistency guarantees for sender and receiver communities.

M65 Convergence rate of optimal quantization and application to the clustering performance of the empirical measure

Y. Liu, G. Pagès (2020). *J. Mach. Learn. Res. 21(86)*

Non-asymptotic convergence rates for optimal quantisers of the empirical measure, i.e. empirical k-means, which give bounds on clustering performance.

M66 Union of low-rank tensor spaces: clustering and completion

M. Ashraphijuo, X. Wang (2020). *J. Mach. Learn. Res. 21(69)*

Deterministic and probabilistic conditions under which data from a union of low-rank tensor subspaces can be clustered and completed.

M67 Community-based group graphical lasso

E. Pircalabelu, G. Claeskens (2020). *J. Mach. Learn. Res. 21(64)*

Estimates communities of variables and a sparse graphical model jointly, with consistency results.

M68 High-dimensional inference for cluster-based graphical models

C. Eisenach, F. Bunea, Y. Ning, C. Dinicu (2020). *J. Mach. Learn. Res. 21(53)*

Variables in clusters are summarised by latent cluster-level variables whose graphical model is the target. After the clusters are estimated, the paper gives asymptotically valid inference on the edges of that model.

M69 Optimal bipartite network clustering [net]

Z. Zhou, A. A. Amini (2020). *J. Mach. Learn. Res. 21(40)*

Bipartite SBMs. Spectral initialisation followed by likelihood refinement attains the minimax rate, including when the two sides are highly unbalanced.

M70 Connecting spectral clustering to maximum margins and level sets

D. P. Hofmeyr (2020). *J. Mach. Learn. Res. 21(18)*

As the scaling parameter vanishes, normalised-cut hyperplanes converge to maximum-margin hyperplanes, and with suitable kernels to low-density (level-set) separators.

M71 Weighted message passing and minimum energy flow for heterogeneous stochastic block models with side information [net]

T. T. Cai, T. Liang, A. Rakhlin (2020). *J. Mach. Learn. Res. 21(11)*

Weighted message passing for heterogeneous SBMs with side information, with recovery thresholds characterised through minimum energy flow on trees.

M72 Path-based spectral clustering: guarantees, robustness to outliers, and fast algorithms [#K]

A. Little, M. Maggioni, J. M. Murphy (2020). *J. Mach. Learn. Res. 21(6)*

Spectral clustering with the longest-leg path distance has finite-sample guarantees for low-dimensional clusters among high-dimensional outliers. The Laplacian eigengap provably identifies the number of clusters, and a quasilinear-time approximation is given.

Selected landmark papers before 2020

M73 DBSCAN: optimal rates for density-based cluster estimation
D. Wang, X. Lu, A. Rinaldo (2019). *J. Mach. Learn. Res. 20(170)*
DBSCAN, suitably tuned, estimates density level-set clusters at minimax-optimal rates.

M74 Community detection and stochastic block models: recent developments [net]
E. Abbe (2018). *J. Mach. Learn. Res. 18(177)*
A survey of the fundamental limits of the SBM (exact recovery, weak recovery and detection thresholds) and of efficient algorithms that achieve them.

M75 Achieving optimal misclassification proportion in stochastic block models [net]
C. Gao, Z. Ma, A. Y. Zhang, H. H. Zhou (2017). *J. Mach. Learn. Res. 18(60)*
Spectral initialisation followed by node-wise likelihood refinement attains the minimax misclassification proportion in polynomial time.

M76 On the estimation of the gradient lines of a density and the consistency of the mean-shift algorithm
E. Arias-Castro, D. Mason, B. Pelletier (2016). *J. Mach. Learn. Res. 17(43)*
Estimated gradient lines (integral curves of the gradient of the kernel density estimate) converge to the population ones, which gives consistency of mean-shift clustering.

M77 Inconsistency of Pitman–Yor process mixtures for the number of components [#K]
J. W. Miller, M. T. Harrison (2014). *J. Mach. Learn. Res. 15, 3333–3370*
Under Dirichlet- and Pitman–Yor-process mixtures, the posterior on the number of clusters is inconsistent for the number of components of a finite mixture.

M78 Stability of density-based clustering
A. Rinaldo, A. Singh, R. Nugent, L. Wasserman (2012). *J. Mach. Learn. Res. 13, 905–948*
Stability of level-set and cluster-tree estimates under resampling and bandwidth variation, with risk bounds that guide the choice of bandwidth.

A Index of results on the number of clusters ([#K])

The table gathers every [#K] entry in the bibliography. It is organised by the *type* of argument used to determine K, and within each group runs from the most recent paper to the oldest. The first column links to the full entry (A = AoS, J = JASA, R = JRSS-B, B = Biometrika, M = JMLR). Landmark papers from before 2020 are included.

Entry	Authors (year)	Jnl	How the number of clusters is determined
1. Hypothesis tests and goodness of fit (used sequentially to select K)			
B2	Liao, Cape (2026)	B	Rank-based Wilcoxon–Wigner matrix, Gaussian edge fluctuations; tests no structure against communities
A9	Zhang, Amini (2023)	A	Adjusted chi-square goodness of fit for DCSBM, $(K+1)$-compression; consistent sequential estimate of K
A10	Han, Yang, Fan (2023)	A	Residual-subsampling rank test with $N(0,1)$ null; consistent sequential rank (= K) estimate
J32	Jin, Ke, Luo, Wang (2023)	J	StGoF: stepwise goodness of fit after MSCORE fitting; optimal estimation of K under DCMM
A20	Yuan, Liu, Feng, Shang (2022)	A	Hypergraph SBM against Erdős–Rényi: detection boundary and count-based tests
A21	Jin, Ke, Luo (2021)	A	Signed-polygon statistics, $K = 1$ against $K > 1$, optimal adaptivity to degree heterogeneity
J40	Hu, Zhang, Qin, Yan, Zhu (2021)	J	Maximum entry-wise deviation of residuals (extreme-value null); sequential testing
B10	Ma, Barnett (2021)	B	Asymptotic normality of modularity in weighted signed networks; test of $K = 1$
A46	Lei (2016)	A	Largest singular value of standardised residual matrix, Tracy–Widom null; sequential
R23	Bickel, Sarkar (2016)	R	Tracy–Widom test of $K = 1$, applied recursively to bipartition
J49	Chen, Li, Fu (2012)	J	EM test for the order of a normal mixture with unequal variances
J50	Li, Chen (2010)	J	EM test of order m_0 against larger orders
A54	Chen, Li (2009)	A	EM test for normal mixtures, tractable limit
J51	Liu, Hayes, Nobel, Marron (2008)	J	SigClust: 2-means cluster index calibrated under a Gaussian null (HDLSS)
R26	Chen, Chen, Kalbfleisch (2001)	R	Modified (penalised) LRT for homogeneity
B17	Lo, Mendell, Rubin (2001)	B	LRT for k_0 against k_1 components, weighted chi-square limit
A58	Dacunha-Castelle, Gassiat (1999)	A	LRT for the order via locally conic parametrisation
A59	Polonik (1995)	A	Excess mass; testing multimodality
A61	Hartigan, Hartigan (1985)	A	Dip test of unimodality
R28	Silverman (1981)	R	Critical-bandwidth test for the number of modes
A63	Hartigan (1978)	A	Asymptotics of the optimal two-cluster split criterion
2. Information criteria, penalised or marginal likelihood			
A2	Li, Peng, Tang, Wu (2026)	A	Information criterion for the number of latent groups in network autoregression
M4	Ren, Zhu, Xu, Ma (2026)	M	Information criterion for the group numbers in multi-relational network autoregression
M8	Chen, Fuh, Kao (2025)	M	Marginal likelihood (parameters and states integrated out); consistent HMM order
J21	Meyer, Wintenberger (2024)	J	Score-type criterion for the number of extremal clusters
J25	Fang, Xu, Xu, Zhu, Guan (2024)	J	Data-driven criterion for the number of groups in network Hawkes processes
R8	Du Roy de Chaumaray, Marbac (2024)	R	Penalised discretised likelihood; selects K and relevant variables
R14	She, Shen, Zhang (2022)	R	Information criterion (non-asymptotic) for the number of feature clusters and the rank
J42	Sugasawa (2021)	J	Information criterion for the numbers of groups and latent components

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Entry	Authors (year)	Jnl	How the number of clusters is determined
A23	Manole, Khalili (2021)	A	Group-Sort-Fuse penalised likelihood on an overfitted mixture
J43	Hu, Qin, Yan, Zhao (2020)	J	Corrected BIC for SBMs
R22	Drton, Plummer (2017)	R	Singular BIC (real log-canonical thresholds)
A43	Wang, Bickel (2017)	A	Likelihood-ratio asymptotics and penalised likelihood for SBMs
J52	Chen, Khalili (2008)	J	Nonsmooth penalty on ordered component differences
A57	James, Priebe, Marchette (2001)	A	Kernel-density versus fitted-mixture discrepancy
A60	Leroux (1992)	A	Penalised likelihood (AIC/BIC-type) does not underestimate K
3. Spectral, rank and eigenvalue methods			
J20	Hwang, Xu, Chatterjee, Bhattacharyya (2024)	J	Number of negative eigenvalues of the Bethe Hessian (sparse SBM/DCSBM)
M12	Du Roy de Chaumaray et al. (2024)	M	Thresholded singular values of an integral operator (HMM order)
B9	Zhang, Jin, Bai (2021)	B	Number of blocks in U-statistic-based matrices
A27	Kwon, Mbakop (2021)	A	Rank of an integral operator for nonparametric mixtures
M48	Dall'Amico, Couillet, Tremblay (2021)	M	Regularised (Bethe-Hessian-type) Laplacian spectrum in sparse graphs
M72	Little, Maggioni, Murphy (2020)	M	Laplacian eigengap with longest-leg path distances
R24	Kasahara, Shimotsu (2014)	R	Rank of matrices of marginal distributions (lower bound on K)
4. Cross-validation, stability and distortion curves			
A4	Jaffe (2025)	A	Strong consistency of the elbow method for k-means-type clustering
B12	Li, Levina, Zhu (2020)	B	Edge cross-validation with low-rank matrix completion
J47	Chen, Lei (2018)	J	Node-pair splitting network cross-validation
B16	Wang (2010)	B	Clustering instability via cross-validation
J53	Sugar, James (2003)	J	Jump method (transformed distortion)
R27	Tibshirani, Walther, Hastie (2001)	R	Gap statistic
5. Bayesian: priors on K and posterior behaviour of the number of clusters			
M10	Yao, Xie, Xu (2025)	M	Sparse high-dimensional mixture with adaptively estimated (growing) K; minimax posterior contraction
B7	Ascolani, Lijoi, Rebaudo, Zanella (2023)	B	Prior on the DP concentration restores consistency for K
M28	Zeng, Miller, Duan (2023)	M	Quasi-Bernoulli stick-breaking: consistent number of clusters
A13	Argiento, De Iorio (2022)	A	Mixtures of finite mixtures: posterior of the numbers of components and clusters
J37	Betancourt, Zanella, Steorts (2022)	J	Microclustering priors: asymptotics of the number of clusters
J44	Xie, Xu (2020)	J	Repulsive priors shrink the posterior on the number of components
J45	Argiento, Cremaschi, Vannucci (2020)	J	Hierarchical NRMIs: prior law of the number of clusters
R20	Fúquene, Steel, Rossell (2019)	R	Non-local priors penalising overlapping or empty components
J46	Geng, Bhattacharya, Pati (2019)	J	MFM-SBM with Bayes-risk bounds for unknown K
J48	Miller, Harrison (2018)	J	Mixtures of finite mixtures (prior on K)
M77	Miller, Harrison (2014)	M	Inconsistency of DP/PY mixtures for the number of components
R25	Rousseau, Mengersen (2011)	R	Overfitted mixtures: small Dirichlet hyperparameter empties extra components
A56	Nobile (2004)	A	Posterior distribution of the number of components
6. Partition recovered without K as input (fusion, sparsity, level sets, recursion)			
J12	Boulin, Di Bernardino, Laloë, Toulemonde (2025)	J	Extremal-dependence variable clustering, exact recovery without K
M15	Chen, Zhang, Ma, Fang (2024)	M	Fusion penalty clusters data sources; clustering consistency
M26	Steinwart, Sriperumbudur, Thomann (2023)	M	All splits of the cluster tree estimated adaptively
J38	Li et al. (2022)	J	Recursive bipartitioning with a stopping rule

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Entry	Authors (year)	Jnl	How the number of clusters is determined
A29	De Castro, Gadat, Marteau, Maugis-Rabusseau (2021)	A	Beurling-LASSO recovers the exact number of mixture atoms
A35	Bing, Bunea, Ning, Wegkamp (2020)	A	LOVE: number of latent clusters estimated consistently
A39	Bunea, Giraud, Luo, Royer, Verzelen (2020)	A	COD variable clustering needs no K
R21	Radchenko, Mukherjee (2017)	R	Post-processed convex-clustering path estimates K
A47	Steinwart (2015)	A	Adaptive estimation of the first split of density level sets

B Ten candidate theoretical projects

These project ideas were suggested by the papers above and are written as research prompts rather than finished plans. Each states the closest existing work, what would be new, a concrete setting, the result to aim for, a possible route and a natural target journal. Projects P1–P4 and P9 concern the number of clusters, components or classes ([#K]). P1 and P2 are about the number of components of a mixture model.

Novelty check

A first draft of ten ideas was checked against the literature on 7 October 2026. The search covered arXiv, the statistics journals, and machine-learning and theoretical-computer-science venues. The outcome:

- Three ideas were **replaced** because their main result already exists or their central conjecture was wrong. They are listed in the table below.
- Six ideas were **kept but refocused** on the part that is still open, because part of their target had already been proved.
- One idea (P10) was **new as stated**.

The check was careful but not exhaustive. Two sub-questions could not be fully searched: clustering under not-at-random missingness (relevant to P6), and BIC/ICL-type or grade-of-membership selection of the number of latent classes (relevant to P9). Both should be rechecked, along with any new preprints, before work starts.

P1 was rechecked on 8 October 2026 against Even, Giraud & Verzelen (2024) and its 2025 sequel, Carpentier & Verzelen (2026) and Royer (2017). As a result, P1 now states which parts those papers already settle: in short, the upper bounds for counting are known at the level of rates, and what remains open is the lower bounds and the moderate-dimensional regime. P3 was sharpened on 9 October 2026 to name the samplers it targets and to rank the possible results.

Original idea	Why it was replaced	Replacement
Selective test for K that does not over-cluster	Wu, Bien & Panigrahi (2025) already estimate K with a guarantee against overestimation, though only on a high-probability event. Split-likelihood "universal inference" (Wasserman, Ramdas & Balakrishnan, 2020) already gives finite-sample control of $\mathbb{P}(\hat{K} > K)$ for i.i.d. mixtures.	P3: single-site Gibbs versus split–merge with random K
Heavy-tailed clustering rates	Jana, Fan & Kulkarni (2025) prove the polynomial rate, with robustness to outliers and adaptivity to the noise law, in fixed dimension. Only sharp constants and the high-dimensional case remain.	P6: clustering Gaussian mixtures with missing entries
Clustering distributions from many small samples	The conjectured consistency with a bounded number of samples per unit is false, because each unit's own Bayes error is bounded below. The corrected lower bound follows from Dreveton et al. (2024).	P9: number of latent classes in degree-heterogeneous models

The projects

P1 Counting versus clustering: when can the number of components of a high-dimensional Gaussian mixture be estimated, and is it easier than finding the clusters? [#K]

Model and notation. We observe $X_1, \dots, X_n \in \mathbb{R}^p$, i.i.d. from

$$P_{\pi,\mu} = \sum_{k=1}^{K} \pi_k \, N_p(\mu_k, I_p),$$

with the noise level known and set to 1. The results sought are non-asymptotic in (n, p, K), and K may grow. Following Carpentier & Verzelen (2026), the regime $n \lesssim pK$ is called *high-dimensional* (HD) and the regime $n \gg pK$ *moderate-dimensional* (MD). Let $\bar{\mu} = \sum_k \pi_k \mu_k$, and let

$$B = \sum_{k=1}^{K} \pi_k (\mu_k - \bar{\mu})(\mu_k - \bar{\mu})^\top = \sum_{k<l} \pi_k \pi_l (\mu_k - \mu_l)(\mu_k - \mu_l)^\top$$

be the between-cluster covariance, so that $\mathrm{Cov}(X_1) = I_p + B$.

- The rank r of B equals the affine dimension of $\{\mu_1, \dots, \mu_K\}$, so $r \le K-1$. Write $\lambda_1 \ge \cdots \ge \lambda_r > 0$ for its non-zero eigenvalues.
- Following Even, Giraud & Verzelen (2024), separation is measured by $\Delta^2 = \min_{k \ne l} \|\mu_k - \mu_l\|^2/2$. In words, the two closest centres are $\sqrt{2}\,\Delta$ apart, in units of the noise standard deviation; with noise variance σ^2, divide by σ^2. In the classes below, Δ is a lower bound on this separation. Thresholds such as $\Delta^2 \asymp \sqrt{pK \log K/n}$ say how large it must be.
- The *simplex configuration* has balanced weights $\pi_k = 1/K$ and centres $\mu_k = \Delta e_k$. It has $r = K-1$ and $\lambda_1 = \cdots = \lambda_{K-1} = \Delta^2/K$.

What is already known.

(K1) *Spiked spectrum (BBP).* Let S be the sample covariance, with K and the λ_j fixed and $p/n \to \gamma < \infty$.

- For $j \le r$, the j-th eigenvalue of S converges to $(1+\lambda_j)(1+\gamma/\lambda_j)$ if $\lambda_j > \sqrt{\gamma}$, and to the bulk edge $(1+\sqrt{\gamma})^2$ otherwise.
- All further eigenvalues converge to the edge.
- When $p/n \to \infty$, the same holds after renormalisation, with threshold $\lambda_j > \sqrt{p/n}$.

References: Baik, Ben Arous & Péché (2005); Baik & Silverstein (2006); Paul (2007); for signal-plus-noise data, Benaych-Georges & Nadakuditi (2012).

(K2) *Spectral counting (fixed K).* Li, Li & Zhu (2024) treat $x_{ij} = \mu_i + \Sigma_0^{1/2} z_{ij}$ with a fixed number τ of groups, fixed proportions and $p/n \to c \in (0, \infty]$.

- They renormalise the centred $n \times n$ Gram matrix.
- The spikes created by the mean differences separate from the bulk exactly when they exceed a critical value. In the isotropic case this is asymptotically the BBP condition $\lambda_j > \sqrt{p/n}$.
- Their estimator is $\hat{\tau} = 1 + \#\{\text{renormalised eigenvalues} \ge 2 + d_n\}$, with $d_n \to 0$ slower than any polynomial. It is consistent when all $\tau - 1$ spikes are above the critical value.

Hence, for fixed K in the simplex configuration, spectral counting works if and only if

$$\Delta^2 > K\sqrt{p/n} = \sqrt{pK^2/n}.$$

(K3) *Non-sharp and growing-K versions.*

- Qing (2026) counts the singular values of the centred $p \times n$ data matrix above $\sqrt{p} + \sqrt{n} + t_n$. This is consistent for $K \le \min(p, n)$ and imbalance $\beta = \min_k n_k/(n/K)$ provided

$$\min_{k \ne l} \|\mu_k - \mu_l\| \;\ge\; 2\sqrt{2}\,\frac{\kappa}{\sqrt{\beta}}\sqrt{\frac{K}{n}}\,(\sqrt{p} + \sqrt{n} + t_n),$$

  where κ is the condition number of the matrix of centres. In the simplex configuration this requires $\Delta^2 \gtrsim Kp/n$ when $p \gg n$, far above the BBP scale $K\sqrt{p/n}$. The paper lists a matching lower bound as open.
- Liu, Liu, Pan, Zhang & Zhang (arXiv:2310.13939) allow the signal rank to grow like $n^{1/3}$. Their eigen-difference criteria for the number of clusters are proved for $p/n \to c < 1$, and are consistent exactly when explicit eigengap conditions hold.
- Classical estimators of the number of spikes use the same spectrum (Kritchman & Nadler, 2009; Passemier & Yao, 2014; Ke, Ma & Lin, 2023).
- When $p \lesssim n$ the rate is elementary. Splitting S into its noise, between-cluster and cross parts gives, with high probability,

$$\|S - (I_p + B)\| \lesssim (1 + \|B\|)\sqrt{p/n} + \|B\|\sqrt{K \log n/n}.$$

  In the simplex configuration, Weyl's inequality then shows that S has exactly $K-1$ eigenvalues above $1 + \Delta^2/(2K)$ once $\Delta^2/K \gg \sqrt{p/n}$ and $n \gg K \log n$. What (K2) adds is the sharp constant. When $p \gg n$ the first term becomes p/n, which leads to Qing's Kp/n scale; the BBP scale there needs random-matrix arguments or (K5).

All of these are *second-moment* methods, so they estimate $r+1$, not K.

(K4) *Clustering with K known.*

- Even, Giraud & Verzelen (2024) treat $p \geq n$ with balanced clusters. Polynomial-time methods (an SDP relaxation of K-means, or single linkage) beat chance when $\Delta^2 \gtrsim \sqrt{pK^2/n} \wedge \sqrt{p \log n}$. Low-degree polynomials fail when $\Delta^2 \ll \sqrt{pK^2/n} \wedge \sqrt{p}$, up to polylogarithmic factors. Exact K-means succeeds when $\Delta^2 \gtrsim \log K \vee \sqrt{pK \log K/n}$, and for $K \geq K_0$ no method succeeds below a constant multiple of this. They point out themselves that the computational barrier sits at the BBP transition; in the simplex configuration it coincides with the spectral-counting threshold of (K2).
- Even, Giraud & Verzelen (2025) prove the same low-degree barrier with a better dependence on the degree and without the restriction $p \geq n$.
- Carpentier & Verzelen (2026) treat MD with $p \geq K$. Low-degree polynomials fail when $\Delta^2 \ll K/n^{1/4} \wedge \sqrt{K}$, up to logarithms; this exceeds the HD barrier once $n \geq (p \vee K)^2$. They conjecture that clustering is possible in polynomial time if and only if $\Delta^2 \gtrsim 1 + \big((\sqrt{pK^2/n} + K/n^{1/4}) \wedge \sqrt{p}\big)$, up to logarithms.
- The gap had been predicted by replica calculations, which give a hard phase once the number of clusters exceeds $4 + 2\sqrt{\alpha}$, where α is the number of samples per dimension (Lesieur et al., 2016). Second-moment bounds also show detection below the spectral threshold when there are many clusters (Banks et al., 2018).

In all of these papers K is an input: every upper bound runs a procedure with K groups, and every lower bound concerns the partition.

(K5) *Counting as a by-product of exact recovery.*

- Royer (2017) replaces the constraint $\mathrm{tr}(Z) = K$ in the SDP relaxation of K-means by a penalty $\hat{\kappa}\,\mathrm{tr}(Z)$. With isotropic noise of known variance, a penalty of order $\sqrt{pn} + n + \sqrt{p \log n}$ is admissible, and the partition, hence K, is recovered exactly with high probability when

$$\Delta^2 \gtrsim (K + \log n) + \sqrt{(K + \log n)\, pK/n}.$$

  When $p \gtrsim n$ and $K \gtrsim \log n$ this is the computational barrier $\sqrt{pK^2/n}$ of (K4).
- Single linkage cut at a distance threshold, rather than at K groups, recovers the partition, hence K, when $\Delta^2 \gtrsim \sqrt{p \log n} + \log n$. This is the argument of Even, Giraud & Verzelen (2024, App. F) with a known noise level.

So when $p \gtrsim n$, polynomial-time counting at the clustering barrier $\sqrt{pK^2/n} \wedge \sqrt{p \log n}$ is already available, up to logarithms. Both methods recover the partition exactly, so they always need $\Delta^2 \gtrsim \log n$.

(K6) *Counting versus finding in other models.* For random graphs with planted communities, with Gaussian or Bernoulli edge weights, testing between two community structures (for example, different numbers of communities) is as hard as recovering the communities, in the low-degree framework (Rush, Skerman, Wein & Yang, 2023). Their Proposition 2.1 bounds the low-degree advantage between two planted distributions for any additive Gaussian model $Y = X + Z$ with an arbitrary prior on X. The Gaussian-mixture data matrix has this form, with rows μ_{z_i} in X, but the bound has not been worked out there.

(K7) *Fixed dimension and higher moments.*

- In fixed dimension the order of a mixture is handled by EM tests (A54, J50, J49), likelihood-ratio asymptotics (A58), penalised likelihood (A60), operator ranks (A27, R24) and the elbow method (A4).
- Without separation, the mixing distribution of a K-component Gaussian location mixture can be estimated in Wasserstein distance at the minimax rate $(p/n)^{1/4} + n^{-1/(4K-2)}$ (Heinrich & Kahn, 2018, in one dimension; Doss, Wu, Yang & Zhou, 2023). The second term comes from higher-order moments.
- Detecting a two-component mixture against a single Gaussian, with sparse mean differences, was analysed by Verzelen & Arias-Castro (2017).

What is new relative to Even, Giraud & Verzelen (2024).

- *Same model, different target.* EGV ask when the partition can be recovered with K given. P1 asks when K itself can be recovered. EGV contain no test or estimator of K; their only mentions of detection refer to Banks et al. (2018).
- *Neither problem implies the other.*

  - A lower bound for recovering a partition gives no lower bound for the binary decision between K and $K+1$.
  - Counting can be much easier than clustering. In MD, spectral counting works once $\Delta^2 \gg \sqrt{pK^2/n}$, by the elementary argument in (K3), and this tends to 0 when $n \gg pK^2$. Clustering needs $\Delta^2 \gtrsim \log K$ even with the centres known: in the simplex configuration the Bayes classifier errs on most points once $\Delta^2 < (2-\delta)\log K$. In a simulation with $K = 30$, $p = 300$, $n = 2 \times 10^5$ and $\Delta^2 = 3.5$, counting the eigenvalues above the bulk edge gave $\hat{K} = 30$ in three runs out of three, while the Bayes classifier with the true centres misclassified 55% of the points.

- *Not new.* Upper bounds for counting at the level of rates. Polynomial-time counting at the clustering barrier, and hence at the BBP rate with growing K, follows from (K5) when $p \gtrsim n$ and from the elementary argument in (K3) when $p \lesssim n$. The coincidence of that barrier with the BBP threshold is pointed out by EGV. Only the sharp constant of spectral counting is open.
- *New.* Lower bounds for counting. The only one that follows from existing results is obtained by splitting a single cluster:

$$\Delta^2_{\text{count}} \gtrsim \sqrt{pK/n}.$$

  Place $K-1$ clusters far away at known positions. Deciding whether the remaining $m \approx n/K$ points form one cluster or two clusters $\sqrt{2}\,\Delta$ apart is symmetric two-component detection, whose information-theoretic and low-degree threshold is $\Delta^2 \asymp \sqrt{p/m}$. (The construction belongs to the classes below when $\alpha \geq 2$ and $c \leq 1/2$.) P1 asks to close the gap between this bound and the upper bounds in HD (Q1, Q2), and to find the counting threshold in MD, where counting is easier than finding (Q3).

What P1 asks. Fix $\alpha \geq 1$ and $c \in (0, 1]$. Let $\mathcal{P}_K(\Delta)$ be the set of K-component mixtures with

- balanced weights, $\max_k \pi_k / \min_k \pi_k \leq \alpha$;
- separated centres, $\min_{k \neq l} \|\mu_k - \mu_l\|^2 \geq 2\Delta^2$;
- a non-degenerate simplex, $\lambda_{K-1}(B) \geq c\,\Delta^2/K$.

Meaning of the third condition.

- K points always lie in an affine subspace of dimension at most $K-1$. They form a non-degenerate $(K-1)$-simplex when this dimension is exactly $K-1$, i.e. when $\mu_2 - \mu_1, \ldots, \mu_K - \mu_1$ are linearly independent; equivalently, $\lambda_{K-1}(B) > 0$. This requires $K - 1 \leq p$.
- The condition makes this quantitative. With balanced weights, $\lambda_j(B) = \sigma_j(\widetilde{M})^2/K$, where $\widetilde{M} = [\mu_1 - \bar{\mu}, \ldots, \mu_K - \bar{\mu}]$. So the condition says $\sigma_{K-1}(\widetilde{M}) \geq \sqrt{c}\,\Delta$: along every unit direction v in the affine hull of the centres, $\sum_k \langle \mu_k - \bar{\mu}, v\rangle^2 \geq c\Delta^2$. In other words, the simplex has no thin direction.
- Examples. The regular simplex has $c = 1$. Random centres, as in the lower-bound constructions of Even, Giraud & Verzelen, are nearly regular when $K \ll p$. Equally spaced collinear centres violate the condition for $K \geq 3$: they have $\lambda_1 = \Delta^2(K^2-1)/6$ and all other $\lambda_j = 0$.
- What it buys. Every component then shows up in the second moments, with a spike of order Δ^2/K. This is what makes the spectral (BBP) benchmark meaningful.

Counting means either testing

$$H_0 : P \in \bigcup_{k \leq K} \mathcal{P}_k(\Delta) \qquad \text{against} \qquad H_1 : P \in \mathcal{P}_{K+1}(\Delta)$$

with vanishing worst-case sum of type I and type II errors, or estimating K consistently, uniformly over $\bigcup_{k \leq K_{\max}} \mathcal{P}_k(\Delta)$. The null is composite: every K-component mixture in the class is allowed.

(Q1) *Information-theoretic threshold for counting.* Find $\Delta^2_{\text{count}}(n, p, K)$ such that counting is possible when $\Delta^2 \gg \Delta^2_{\text{count}}$ and impossible when $\Delta^2 \ll \Delta^2_{\text{count}}$.

- Known: $\sqrt{pK/n} \lesssim \Delta^2_{\text{count}} \lesssim \sqrt{pK^2/n} \wedge \sqrt{p \log n}$, by the split reduction, (K5) when $p \gtrsim n$, and (K3) when $p \lesssim n$.
- Conjecture: $\Delta^2_{\text{count}} \asymp \min\{\sqrt{pK^2/n},\ \log K \vee \sqrt{pK \log K/n}\}$. That is, one can count either spectrally or by clustering, and nothing else helps. In HD the minimum is the clustering threshold, so there would be no counting–finding gap at the information-theoretic level. Either answer is informative: if the split bound $\sqrt{pK/n}$ is attained instead, counting is information-theoretically easier than finding by a factor $\sqrt{\log K}$.
- Upper bound: exact K-means with a penalty on the number of groups, as in Royer's SDP, analysed with EGV's arguments. This should be routine.

- Lower bound: the main difficulty. The null is composite, so it needs priors on *both* hypotheses whose induced data distributions are close (fuzzy hypotheses). It must also gain the factor $\sqrt{\log K}$ over the split reduction, which Fano's method gives for partitions but not for a binary decision.

(Q2) *Computational threshold for counting in HD.* Show that low-degree tests of K against $K+1$ fail when $\Delta^2 \ll \sqrt{pK^2/n} \wedge \sqrt{p}$, up to polylogarithmic factors, for K larger than a power of $\log n$.

- With (K5), this would pin down the computational threshold of counting and make it equal to that of clustering. Counting would be as hard as finding, as (K6) shows for graphs.
- Together with (Q1), it gives a computation–information gap for counting in the window $\sqrt{pK \log K/n} \ll \Delta^2 \ll \sqrt{pK^2/n}$.
- Route: apply the planted-versus-planted bound of (K6) to two priors (random simplices with K and $K+1$ vertices), and control the resulting cumulant sums with the scheme of Even, Giraud & Verzelen (2025).
- The two simplex priors differ in low-order spectral moments, e.g. $\|B\|_F^2 = \Delta^4(K-1)/K^2$ against $\Delta^4 K/(K+1)^2$. The difference, about Δ^4/K^2, is visible to degree-4 statistics only when $\Delta^2 \gtrsim K\sqrt{p/n}$. So these naive priors should suffice up to constants; moment-matched priors would be needed only for sharp constants.

(Q3) *Moderate dimension: counting is easier than finding.* When $n \gg pK$, clustering needs $\Delta^2 \gtrsim \log K$, and in polynomial time conjecturally $\Delta^2 \gtrsim 1 + \big((\sqrt{pK^2/n} + K/n^{1/4}) \wedge \sqrt{p}\big)$ (K4). Counting needs much less.

- Upper bound: at the level of rates this is the elementary argument in (K3). What remains is the sharp constant: consistency when $\lambda_{K-1} \geq (1+\epsilon)\sqrt{p/n}$, with a data-driven threshold, $K \to \infty$ and unequal weights, extending Li, Li & Zhu beyond fixed K.
- Lower bound (the real question): show that $\Delta^2_{\text{count}} \gtrsim \sqrt{pK^2/n}$ whenever this is below $\log K$, so that spectral counting is minimax there. Only $\sqrt{pK/n}$ is known. For spiked covariance models with a fixed number of spikes, the analogous statement is contiguity below the BBP threshold (Onatski, Moreira & Hallin, 2013, 2014). Here the spikes come from a discrete mixture, and their number grows.
- Together these would give a counting–finding gap between $\sqrt{pK^2/n}$ and $\log K$, which is unbounded.

(Q4) *Light components (secondary).* Let the extra component have weight $\varepsilon \to 0$ and lie at distance D from the other centres. It can be detected

- spectrally, when its spike $\approx \varepsilon D^2$ beats $\sqrt{p/n}$, i.e. $D^2 \gtrsim \varepsilon^{-1}\sqrt{p/n}$;
- by flagging single outlying points, when $D^2 \gtrsim \sqrt{p \log n}$;
- by an exhaustive scan over subsets of size εn, when $D^2 \gtrsim \sqrt{p \log(1/\varepsilon)/(\varepsilon n)} + \log(1/\varepsilon)$.

Against a one-cluster null these rates are known. With a random centre direction, the p columns of the data matrix are i.i.d. $N_n\big(0, I_n + (D^2/p)\mathbf{1}_S\mathbf{1}_S^\top\big)$ with $|S| = \varepsilon n$. This is sparse PCA in dimension n, with p samples and sparsity εn (Berthet & Rigollet, 2013; Ding, Kunisky, Wein & Bandeira, 2024). Only the composite K-component null is new: the aim is to show that the scan rate stays minimax and the low-degree barrier $\min\{\varepsilon^{-1}\sqrt{p/n}, \sqrt{p\log n}\}$ persists.

(Q5) *Without the non-degeneracy condition.* Drop the third condition and keep only balance and separation.

- *Second-moment methods break down.* They estimate one plus the number of spikes of B above the BBP level, which is at most $1+r$, where r is the affine dimension. For $K \geq 3$ equally spaced collinear centres they return at most 2, however large Δ is.
- *High dimension ($p \gtrsim n$): degeneracy should not make counting harder.* Separation at the conjectured threshold is many noise standard deviations. A low-dimensional configuration also concentrates its signal in a few directions that are easy to see. Collinear centres, for example, have $\lambda_1 = \Delta^2(K^2-1)/6 \gg \sqrt{p/n}$ there, so one can project onto the estimated line and count well-separated clusters in one dimension. The conjecture is that the thresholds of (Q1) and (Q2) still hold, attained by an adaptive procedure: estimate the visible subspace, then count clusters inside it by clustering or moment methods.
- *Low dimension: degeneracy changes the rate.* Heuristically, take p and K fixed. If the $K+1$ centres of the alternative span K dimensions, the extra component is detected through the K-th eigenvalue of the covariance once $\Delta \gg n^{-1/4}$. If they are collinear, a K-atom and a

$(K+1)$-atom configuration on the line can share all moments up to order $2K-1$ (but not $2K$). The chi-square distance between the two mixtures is then of order Δ^{4K}, so $\Delta \gg n^{-1/(4K)}$ is needed (compare (K7)).

- *Aim.* Establish this transition uniformly in the affine dimension r, together with the matching computational limits of projection and tensor methods.

Task	Information-theoretic	Polynomial time	Status
Clustering, K known	$\log K \vee \sqrt{pK \log K/n}$	HD: $\sqrt{pK^2/n} \wedge \sqrt{p}$ MD: $1 + ((\sqrt{pK^2/n} + K/n^{1/4}) \wedge \sqrt{p})$	known (K4), except the MD upper bound (conjectured)
Counting, known bounds	from $\sqrt{pK/n}$ to $\sqrt{pK^2/n} \wedge \sqrt{p \log n}$	same	lower: split reduction; upper: (K5) for $p \gtrsim n$, (K3) for $p \lesssim n$
Counting, conjecture	$\min\{\sqrt{pK^2/n},\ \log K \vee \sqrt{pK \log K/n}\}$	$\sqrt{pK^2/n} \wedge \sqrt{p}$	open (Q1–Q3)

Thresholds on $\Delta^2 = \min_{k \neq l} \|\mu_k - \mu_l\|^2/2$ in the simplex configuration, up to constants and logarithmic factors. HD: $n \lesssim pK$; MD: $n \gg pK$.

Route.

- Q1: second-moment bounds for fuzzy hypotheses, with priors on the centres under both hypotheses; for the upper bound, EGV's analysis of exact K-means with a penalty on the number of groups.
- Q2: the planted-versus-planted bound of Rush et al. (their Proposition 2.1), with the cumulant scheme of Even, Giraud & Verzelen (2025).
- Q3: resolvent methods or isotropic local laws for spiked matrices of growing rank (Li, Li & Zhu; Liu et al.) for the upper bound; second-moment bounds for randomly rotated simplices, in the spirit of Onatski, Moreira & Hallin, for the lower bound.
- Q4: sparse-PCA lower bounds, adapted to composite nulls.
- Q5: moment and tensor methods with local entropy bounds, as in Doss et al.

Venue. AoS; Q2 on its own would also suit COLT.

P2 What is the number of components under misspecification? A tempering window for the posterior of K [#K]

Prior work.

- The posterior of K is consistent for well-specified mixtures (J48, A13, B7, M28).
- Under kernel misspecification it diverges (Cai, Campbell & Broderick, 2021). This also holds for power posteriors whose power is fixed or converges to a positive constant (Cai, Campbell & Broderick, 2020).
- Coarsening corresponds to a power posterior with $n\zeta_n = O(1)$ (Miller & Dunson, 2019).
- A penalised *frequentist* criterion with a component-level tolerance is consistent for K under misspecification (Li & Huggins, 2024).
- Merge-based and dendrogram methods are consistent only under well-specification (Guha, Ho & Nguyen, 2021; Do et al., 2024). FOLD (J9) targets an oracle clustering, not the posterior of K.

What is new. The intermediate regime $\zeta_n \to 0$ with $n\zeta_n \to \infty$, and a posterior (not frequentist) definition of "the number of clusters", with sharp bounds.

Setting. $P_0 = \sum_{j \leq m} w_j F_j$ is a mixture of well-separated non-Gaussian components, fitted by a Gaussian mixture of finite mixtures. The analysis uses the fractional posterior $\Pi_\zeta \propto \prod_i p_\theta(X_i)^\zeta \Pi$.

Target. Let g_{sep} be the gain in KL divergence from separating the true clusters, and g_{split} the gain from splitting one cluster into extra Gaussians. A Laplace heuristic suggests that $\Pi_{\zeta_n}(K = m \mid X) \to 1$ exactly in the window $g_{\text{sep}}^{-1} \log(1/g_{\text{sep}}) \lesssim n\zeta_n \lesssim g_{\text{split}}^{-1} \log(1/g_{\text{split}})$, which is non-empty only when $g_{\text{sep}}/g_{\text{split}}$ is large. The aim is to prove this non-asymptotically, with matching lower bounds, and so to define the number of clusters of a non-mixture distribution as the location of a gap in its KL-gain profile.

Route. Misspecified posterior contraction (Kleijn & van der Vaart, 2006); fractional posteriors (Bhattacharya, Pati & Yang, 2019); partition calculus for mixtures of finite mixtures (A13); identifiability through clustering of mixture atoms (A34).

Venue. AoS or JRSS-B.

P3 Single-site Gibbs versus split–merge: quantitative MCMC convergence for Bayesian clustering with a random number of components [#K]

Why this question. When K is random, the usual samplers are known to converge eventually. This is trivial for collapsed samplers on partitions, whose state space is finite; for reversible jump it holds by construction, with the fine print of Roberts & Rosenthal (2006). Nothing quantitative is known. In practice, single-site samplers are seen to rarely change the number of clusters, and split–merge moves are added to fix this (Jain & Neal, 2004; Peixoto, 2020, for SBMs). No theorem explains when this is necessary.

Which sampler. The most useful target is the collapsed samplers over partitions, because they are what people run:

(S1) single-site collapsed Gibbs with a "new cluster" option, for the posterior of a mixture of finite mixtures (Miller & Harrison's version of Neal's algorithm, J48), and its Dirichlet-process counterpart;

(S2) (S1) plus split–merge Metropolis–Hastings moves (Jain & Neal, 2004).

In these samplers K changes only when a cluster is created or emptied, which is exactly where the difficulty of random K lies. Reversible jump (Richardson & Green, 1997) and birth–death samplers (Stephens, 2000) carry continuous parameters and Jacobians, are harder to analyse and are less used today. The telescoping sampler (Frühwirth-Schnatter, Malsiner-Walli & Grün, 2021) is a natural second target.

Setting. A mixture-of-finite-mixtures prior: $K \sim p_K$ (Poisson, geometric or beta-negative-binomial) and symmetric Dirichlet(γ) weights. Conjugate likelihoods, so that the posterior lives on partitions of $\{1, \dots, n\}$:

(a) an isotropic Gaussian mixture with centres $\mu_k \sim N(0, \tau^2 I_p)$ and known noise level;

(b) an SBM with Beta priors on the block probabilities (as in J46).

The data come from a balanced, well-separated truth with K_0 clusters. The analysis is restricted to the regime where the posterior concentrates on the true partition, so that the question is purely computational.

Prior work.

- Rapid mixing has been proved for Metropolis–Hastings in the SBM with K known, from a warm start (M60).
- For Gaussian mixtures with k fixed there are only lower bounds for collapsed Gibbs, which are exponential when k is wrong (Tosh & Dasgupta, 2014).
- Polynomial mixing is known for one Metropolis-type sampler, but only for the tempered (power) posterior of a symmetric two-component mixture (Mou, Ho, Wainwright, Bartlett & Jordan, 2019).
- A fast non-reversible sampler, with scaling limits, exists for fixed K; random K is explicitly left open (Ascolani & Zanella, 2025). Newman (2025) gives a fast sampler for unknown K, with empirical evidence only.
- Coupled and sandwich samplers come with qualitative guarantees only (Hobert, Roy & Robert, 2011; Nguyen, Trippe & Broderick, 2022).

Targets, in decreasing order of interest.

(i) *A separation theorem: (S1) is slow and (S2) is fast.*

- High-dimensional heuristic, case (a). A new cluster starts as a single point. That point carries signal of order Δ^2 against noise of order p about the new centre, so (S1) should essentially never open a missing cluster unless $\Delta^2 \gtrsim p$, up to logarithms and prior-dependent constants.
- Yet the posterior should already concentrate on the true K once $\Delta^2 \gtrsim (pK/n) \log n$, since the collapsed posterior carries a BIC-like Occam factor.
- So in the window $(pK/n) \log n \ll \Delta^2 \ll p$, (S1) started with too few clusters should stay stuck for exponentially long, while (S2) should reach the right K in polynomially many steps.
- In fixed dimension the barrier to opening clusters disappears. The comparison then becomes $O(n^2)$ single-site updates to merge a spuriously split cluster, against $O(1)$ split–merge proposals.

(ii) *Hitting time of the correct K* from standard starting points: all points in one cluster, each point in its own cluster, or K-means with a wrong K. This is what practitioners monitor, and it may be easier to prove than full mixing.

(iii) *Local mixing from a warm start with random K.* Extend M60 from known K to the mixture-of-finite-mixtures posterior with create and empty moves. This shows that moves changing K do not destroy rapid mixing near the truth.

(iv) *The scaling limit of Ascolani & Zanella for random K.*

What would count as interesting.

- Any polynomial bound, for a sampler people actually use, with explicit dependence on n, K, Δ and p. None exists for random K.
- Target (i) is the strongest. It would turn folklore into a theorem with a clear practical message: when split–merge moves are necessary.
- Targets (ii) and (iii) in the SBM make a reasonable first paper, because M60 gives a template.
- Not interesting: bounds exponential in n with no matching lower bound, which finite-state theory gives for free; samplers designed only to be analysable; and regimes where the posterior is so concentrated that mixing is trivial.

Venue. AoS or JRSS-B for (i); Ann. Appl. Probab., Bernoulli or COLT for (ii)–(iii). Ascolani and Zanella list random K as future work, so this would need to move quickly.

P4 How many communities does a sparse multilayer network have? [net] [#K]

Multilayer SBM. Fix n nodes, L layers and K communities. A multilayer stochastic block model is specified by

- a label vector $z \in [K]^n$ in which every community is non-empty, with membership matrix $Z \in \{0,1\}^{n\times K}$, $Z_{ik} = \mathbf{1}\{z_i = k\}$, and community sizes $n_k = \#\{i : z_i = k\}$;
- symmetric connectivity matrices $B^{(1)}, \dots, B^{(L)} \in [0,1]^{K\times K}$, one per layer.

We observe symmetric adjacency matrices $A^{(1)}, \dots, A^{(L)} \in \{0,1\}^{n\times n}$ with zero diagonal. Given z, the entries $\{A_{ij}^{(\ell)} : i < j,\ 1 \le \ell \le L\}$ are independent, with

$$A_{ij}^{(\ell)} \sim \text{Bernoulli}\big(B^{(\ell)}_{z_i z_j}\big).$$

So all layers share the communities, but each layer has its own connection probabilities. Off the diagonal, $\mathbb{E}A^{(\ell)} = P^{(\ell)} := ZB^{(\ell)}Z^\top$. With $L = 1$ this is the ordinary SBM. For K to be well defined, no two communities may have the same connection probabilities in every layer; that is, the rows of the $K \times KL$ matrix $[B^{(1)} \ \cdots \ B^{(L)}]$ are distinct.

Sparse multilayer SBM. Write $B^{(\ell)} = \rho_n B_0^{(\ell)}$, where $\rho_n \in (0,1]$ is the edge density and $B_0^{(\ell)} \in [0,1]^{K\times K}$, and let the number of layers $L = L_n$ grow with n. Expected degrees in each layer are then of order $n\rho_n$.

- *Sparse* means $n\rho_n = O(1)$, or even $n\rho_n \to 0$. A single layer is then useless for finding the communities, since most nodes have bounded or zero degree. Only the layers together carry the signal.
- *Summing the layers can fail.* The mean of $\sum_\ell A^{(\ell)}$ is $\rho_n Z\big(\sum_\ell B_0^{(\ell)}\big)Z^\top$, and $\sum_\ell B_0^{(\ell)}$ can lose rank through cancellation. For $K = 2$, the layers $B_0^{(1)} = \left(\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right)$ and $B_0^{(2)} = \left(\begin{smallmatrix} b & a \\ a & b \end{smallmatrix}\right)$ sum to $(a+b)\mathbf{1}\mathbf{1}^\top$, which has no community structure.
- *Squaring cannot cancel.* The bias-adjusted sum of squares (J30) is

$$S = \sum_{\ell=1}^{L} \Big\{ \big(A^{(\ell)}\big)^2 - D^{(\ell)} \Big\}, \qquad D^{(\ell)} = \operatorname{diag}\big(A^{(\ell)}\mathbf{1}\big).$$

  For $i \neq j$, S_{ij} counts the paths of length two from i to j in all layers, and the diagonal of S is zero. Off the diagonal, $\mathbb{E}S \approx Q := \sum_\ell \big(P^{(\ell)}\big)^2$, a matrix of rank K. With balanced communities ($n_k \approx n/K$) its non-zero eigenvalues are

$$\mu_j \approx \frac{n^2\rho_n^2}{K^2}\,\lambda_j\Big(\sum_{\ell=1}^{L}\big(B_0^{(\ell)}\big)^2\Big), \qquad j = 1, \dots, K,$$

  and $\sum_\ell \big(B_0^{(\ell)}\big)^2$ is positive definite as soon as one layer has full rank.

Let $\mathcal{M}_K = \mathcal{M}_K(n, L, \rho_n; \alpha, c_0)$ be the set of sparse multilayer SBMs with K communities such that

- the communities are balanced, $n/(\alpha K) \le n_k \le \alpha n/K$;
- the layers are jointly non-degenerate, $\lambda_K\big(\sum_\ell (B_0^{(\ell)})^2\big) \ge c_0 L$, so that $\mu_K \gtrsim c_0 L\, n^2\rho_n^2/K^2$.

The regime of interest is $n\rho_n = O(1)$, or $n\rho_n \to 0$, with $n\rho_n\sqrt{L} \to \infty$. In this regime the bias-adjusted spectral clustering of J30 recovers the communities when K is known.

What is already known.

(K1) *Single networks.* Estimating K is well understood (A46, A10, J32), including sparse graphs, through the Bethe Hessian (J20).

(K2) *Dense multilayer networks.* Sequential goodness-of-fit tests on a normalised aggregate of the residuals estimate K consistently. Each layer may have its own $B^{(\ell)}$, and L may grow, roughly up to $o(n/\log n)$ (Qing, 2025, 2026). The theory requires every connection probability to lie in $[\delta, 1-\delta]$.

(K3) *Sparse multilayer networks with K known.* Least squares (B13), bias-adjusted sums of squares (J30), spectral and matrix-factorisation methods (A37), and pruned sums of squares (Bhattacharyya & Chatterjee, manuscript).

(K4) *Thresholds.* With two blocks, unrestricted methods recover the communities once $n\rho_n L$ is large enough, but low-degree polynomials need $n\rho_n\sqrt{L}$ to be large (A5).

(K5) *The easy case.* If the $B_0^{(\ell)}$ do not cancel, $\sum_\ell A^{(\ell)}$ behaves like a single weighted SBM, and single-network estimators of K should carry over. No written proof was found, but this is likely routine.

What P4 asks. Estimate K uniformly over $\bigcup_{k\le K_{\max}} \mathcal{M}_k$, or test $H_0 : \mathcal{M}_K$ against $H_1 : \mathcal{M}_{K+1}$, in the sparse regime.

(Q1) *A sharp noise edge for S.* Find a deterministic r_n, depending on n, ρ_n, L and the variance profile of S, such that, uniformly over $\mathcal{M}_K$:

- $\lambda_{K+1}(S) \le (1+o(1))\, r_n$ with high probability, so the noise creates no spurious outliers;
- there is a BBP-type transition, as for a deformed Wigner matrix: if $\mu_K \ge (1+\epsilon)r_n/2$ then $\lambda_K(S) \ge (1+c_\epsilon)r_n$, while if $\mu_K \le (1-\epsilon)r_n/2$ then $\lambda_K(S) \le (1+o(1))\, r_n$.

Heuristically, $S - \mathbb{E}S$ is a sum of L independent centred matrices, so it should behave like a Wigner matrix with a variance profile. If all connection probabilities are about $b\rho_n$, this gives $r_n \approx 2b\sqrt{L}\, n\rho_n$, with the transition at $n\rho_n\sqrt{L} \approx bK^2/c_0$. A quick check with Erdős–Rényi layers agrees: for $n = 2000$, $n\rho_n = 1.5$ and $L = 400$, the largest eigenvalue of $S - \mathbb{E}S$ is 62.1, against $2\sqrt{L}\, n\rho_n = 60$.

The obstacle is high-degree vertices. A vertex of degree d in one layer turns its neighbours into a clique of 2-paths, which gives S an eigenvalue of about $d-1$. When $n\rho_n = O(1)$ the maximum degree is about $\log n/\log\log n$, which exceeds r_n unless L is large. In the same check with $n = 3000$, $n\rho_n = 1$ and $L = 4$, the largest eigenvalue is 6.9, close to the maximum degree 7 and well above $2\sqrt{L}\, n\rho_n = 4$. This may call for pruning high-degree vertices, as Bhattacharyya & Chatterjee do, or for a non-backtracking analogue of S.

(Q2) *A data-driven estimator.* $\hat{K} = \#\{j : \lambda_j(S) > (1+\eta)\hat{r}_n\}$, with $\hat{r}_n$ estimated from the data and η small. It should be consistent without extra logarithmic factors in the signal condition.

(Q3) *Lower bounds for testing K against K + 1, for general K.* An information-theoretic bound at the scale $n\rho_n L$ and a low-degree bound at the scale $n\rho_n\sqrt{L}$, up to factors depending on K. This would extend A5 from recovery with two blocks to counting, and show whether counting has the same $\sqrt{L}$ gap.

Venue. AoS or JASA.

P5 Honest and adaptive confidence sets for a partition

Prior work.

- Joint confidence sets for group membership, obtained by inverting unit-wise tests, exist for panel data (Dzemski & Okui, 2024).
- Confidence sets for a two-community SBM can be obtained by enlarging credible sets (Kleijn & van Waaij, 2025).
- Conformal label sets have only per-point coverage (Hur, Nath & Allen, 2026).

- Pairwise membership tests (R15) and equivalence tests (A19) do not give sets for the whole partition.
- None of these work characterises the size of the sets.

What is new. Minimax and adaptivity theory for confidence sets of a partition.

Setting. An isotropic K-component Gaussian mixture or an SBM, with separation Δ. The loss is the Hamming distance up to relabelling.

Target. Three questions.

(a) What is the minimax diameter of honest $(1-\alpha)$ sets over $\{\Delta \geq \Delta_0\}$? Is it of the order of the estimation rate, $n\exp\{-(1+o(1))\Delta_0^2/8\}$?

(b) Can one procedure attain it for every Δ at once? The conjecture is no in general, by analogy with nonparametric confidence balls (Cai & Low, 2006; Robins & van der Vaart, 2006), but yes for $\Delta \gtrsim \sqrt{\log n}$ or under a self-similarity condition.

(c) Can non-asymptotic constructions from leave-one-out membership scores meet these bounds? The constructions of Dzemski–Okui and Kleijn–van Waaij would serve as baselines.

Venue. AoS.

P6 Clustering Gaussian mixtures with missing entries: sharp misclustering exponents

Prior work.

- For categorical data with missing entries, exact recovery is known up to polylogarithmic factors (J17).
- Subspace estimation and heteroskedastic PCA with missing data supply the spectral tools (Cai, Li, Chi, Poor & Chen, 2021; Zhang, Cai & Wu, 2022; Yan, Chen & Fan, 2024).
- Under MCAR, k-means centres are $\sqrt{n}$-consistent (Guan, 2026), but k-POD is inconsistent (Terada & Guan, 2024).
- The supervised versions (LDA and classification with missing data) have minimax theory (Cai & Zhang, 2019; Sell, Berrett & Cannings, 2024).
- No misclustering rates for Gaussian mixtures with missing coordinates were found.

What is new. Sharp misclustering exponents and exact-recovery thresholds for unsupervised clustering with missing coordinates.

Setting. An isotropic K-component Gaussian mixture in which each coordinate is observed independently with probability q (MCAR), or with probability depending on the class (MNAR).

Target.

(i) The minimax misclustering proportion equals the oracle rate $\exp\{-(1+o(1))I_q\}$, where $I_q = \min_{a\neq b}\sum_j -\log\{1-q+q\,e^{-\delta_{ab,j}^2/(8\sigma^2)}\}$. This is about $q\Delta^2/(8\sigma^2)$ when the mean differences δ_{ab} are spread over many coordinates.

(ii) The rate is attained by spectral initialisation on the diagonal-deleted Gram matrix followed by refinement on the observed coordinates, whereas Lloyd's algorithm on zero-imputed data is suboptimal.

(iii) Exact-recovery thresholds.

(iv) Under class-dependent missingness, the exponent gains a Chernoff term between the distributions of the missingness patterns.

Venue. AoS or JASA.

P7 Dynamic stochastic block models with switching memberships: lower bounds and the phase diagram [net]

Prior work.

- Kernel-smoothed, debiased sum-of-squares clustering (B1) has error of order $sh + \log n/\{h(n\rho)^2\}$, where s is the switching probability per step and h the window. Hence it is consistent once $n\rho\sqrt{w} \gg \sqrt{\log n}$, where $w = \min(T, 1/s)$ is the memory length. That paper proves no lower bounds.
- For a constant, known connectivity matrix, belief propagation reaches the cavity-method detectability threshold, so there is no gap (Ghasemian et al., 2016).

- Sharp weak-recovery thresholds at large degree have been derived under unproven assumptions (Yang, Lin & Sen, 2024).
- Static multilayer thresholds are known (A5).

What is new. Matching lower bounds, which would establish the full phase diagram with switching memberships.

Setting. T snapshots, density ρ, and Markov switching with probability s. The connectivity matrices $B^{(t)}$ are heterophilic or unknown, the case where a gap is expected.

Target.

(i) An information-theoretic threshold at the scale $n\rho w$, proved in both directions.

(ii) Low-degree hardness below $n\rho\sqrt{w}$, showing that the existing upper bound is computationally optimal.

(iii) Chernoff–Hellinger constants for exact recovery under node-wise Markov switching.

(iv) A bandwidth choice with guarantees that adapts to unknown s.

Venue. AoS or JRSS-B.

P8 The cost of privacy in Gaussian-mixture clustering: joint versus local differential privacy

Prior work.

- For SBMs, the order of the privacy cost of recovery is known under edge DP (Chen et al., 2023; Mohamed, Nguyen, Vullikanti & Tandon, 2022; Nguyen & Vullikanti, 2024) and under node DP (Klopp & Zadik, 2026), but only up to constants.
- For Gaussian mixtures, private algorithms learn the parameters or the density (Kamath, Sheffet, Singhal & Ullman, 2019; Afzali, Ashtiani & Liaw, 2023; Chen et al., 2023), but give no misclustering rates.
- The cost-of-privacy programme (Cai, Wang & Zhang, 2021) has not reached clustering.

What is new. Sharp misclustering theory for Gaussian mixtures under privacy, and the contrast between joint and local DP.

Setting.

- Joint DP: centres are released under central (ε, δ)-DP, and each individual computes their own label.
- Local DP: each individual privatises their own data.

Target. Minimax misclustering rates, and private separation and sample-size conditions with matching lower bounds, as functions of $(n, p, K, \Delta, \varepsilon, \delta)$. The conjecture is that under joint DP the exponent $\Delta^2/8$ survives once the error of the private centres is $o(\Delta)$, so privacy costs only samples and separation, whereas under local DP the exponent itself degrades. A constant-sharp edge-DP exact-recovery threshold for the two-community SBM would be a secondary target.

Venue. AoS or JASA.

P9 How many latent classes? Sharp estimation of K in degree-heterogeneous high-dimensional latent class models [#K]

Prior work.

- For the standard binary latent class model, K can be estimated consistently by thresholding singular values at a worst-case level of about $2(\sqrt{J} + \sqrt{N})$, with K fixed and no lower bound (Lyu & Gu, 2026).
- Sequential goodness-of-fit tests estimate K for ordinal data, but require $JK\log(JK)/N \to 0$ and assume no degree heterogeneity (Qing, 2026).
- The degree-heterogeneous model of J5 takes K as known.
- The network analogue, with severe degree heterogeneity, is solved in J32.

What is new. Sharp thresholds and lower bounds for estimating K under degree heterogeneity, with J of the order of N or larger.

Setting. N subjects and J items with $J/N \to c \in (0, \infty]$. The model is the degree-heterogeneous latent class model, and K may grow.

Target.

(i) A normalised, degree-adjusted residual statistic with a sharp, parameter-free null limit, which

yields a stepwise estimator of K that is consistent above an explicit signal-to-noise threshold.

(ii) A matching lower bound below which no estimator of K is consistent.

(iii) An extension to polytomous items.

Venue. JASA or AoS.

P10 Provable clustering on self-supervised representations

Prior work.

- Spectral contrastive learning has accuracy guarantees for a *supervised linear probe* (HaoChen, Wei, Gaidon & Ma, 2021).
- InfoNCE minimisers preserve clusters at the population level (Parulekar et al., 2023) and are equivalent to spectral clustering on a similarity graph (Tan, Zhang, Yang & Yuan, 2024).
- Limited model capacity recovers special clustering structures (HaoChen & Ma, 2023).
- InfoNCE finds the optimal subspace in Gaussian mixtures (Bansal, Kavis & Sanghavi, 2024).
- The augmentation-graph Laplacian is consistent and realisable (Li & Neuman, 2025).
- Theory for fixed embeddings also exists (M40, M55).

What is new. Unsupervised misclustering guarantees, rather than linear-probe guarantees, together with lower bounds and the choice of K.

Setting. Data $x = g(z, u)$ come from a latent class $z \in [K]$ and noise u, with an augmentation kernel that preserves z. Features are learned from n unlabelled points by a function class $\mathcal{F}$.

Target.

(i) A misclustering bound for k-means on the learned features, of the form eigengap or conductance term plus a complexity term for $\mathcal{F}$.

(ii) A lower bound showing that the eigengap condition is necessary.

(iii) A provably consistent rule for choosing K from the spectrum of the learned features.

The first is the expected route through a structure theorem. The second and third are the clearly new parts.

Venue. JMLR or AoS.

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